

Factivity, presupposition projection, and the role of discrete knowledge in gradient inference judgments

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Abstract

We investigate whether the factive presuppositions associated with some clause-embedding predicates are fundamentally discrete in nature—as classically assumed—or fundamentally gradient—consistent with recent proposals that projection is gradient in general (e.g., the Gradient Projection Principle of Tonhauser, Beaver, and Degen 2018). To carry out this investigation, we develop statistical models of presupposition projection that implement these two hypotheses, fit these models to existing inference judgment data aimed at measuring projection (Degen and Tonhauser 2021), and compare the models’ fits to the data using standard statistical model comparison metrics. We find that models implementing the hypothesis that presupposition projection is fundamentally discrete fit the data better than models implementing the hypothesis that it is fundamentally gradient. To evaluate the robustness of this finding, we collect three additional datasets: a replication of the original dataset, as well as two datasets that modify the methodology of the original. Across the three datasets, we again find that models implementing the hypothesis that presuppositions are discrete fit the data better than models implementing the hypothesis that they are gradient. We argue that these results favor an account according to which factive presuppositions are fundamentally discrete in nature, and we discuss how this discreteness might be understood within classical semantic accounts of factive predicates, as well as accounts of presupposition projection that tie it to the question under discussion.

Keywords: factivity, presupposition, projection, probabilistic semantics, gradience

1 Introduction

Semantic theories aim to characterize the inferences that natural language expressions support and to account for at least those inferences which are necessary given the meanings of the expressions. Whether or not a particular inference is necessary is commonly assessed via native speaker judgments. Judgment data, however, tends to be influenced by many non-semantic factors. These factors run the gamut: from high-level factors, such as (a) speakers' prior beliefs about the likelihood that an inference is true or (b) syntactic and semantic ambiguities, to low-level factors, such as (a) the strategies comprehenders use to map their judgments onto a data collection instrument (e.g., a slider representing likelihood or certainty) and (b) their skill in producing an accurate target response using such an instrument.¹

Testing a semantic theory against inference judgment data therefore requires auxiliary assumptions about the link between (some representation of) these factors and the theoretical constructs of interest. Such linking assumptions are often left implicit in classical methodologies employing informal experiments.² In recent years, however, the need to formulate explicit linking assumptions has become pressing in light of theoretical developments within semantics that have been motivated by finer-grained aspects of the distribution of inference judgments than can be observed informally.

One domain where large datasets of inference judgment have become important is presupposition projection and, in particular, factivity. A predicate is said to be *factive* if it is implicated in triggering *veridicality inferences* (i.e., inferences that its embedded clause is true) regardless of whether or not certain entailment canceling operators take scope over the predicate (Kiparsky and Kiparsky 1970). For example, *love* is often taken to be factive, since sentences such as those in (1) give rise to the inference that (2).

- (1) a. Jo loves that Mo left.
b. Jo doesn't love that Mo left.
c. Does Jo love that Mo left?
d. Jo might love that Mo left.
e. If Jo loves that Mo left, she'll also love that Bo left.

- (2) Mo left.

The inference that (2) is thus said to *project* through all entailment cancelling operators in (1). Diagnosing factivity based on inference judgments is known to be

¹We refer to as “high-level” those factors which we take to underlie linguistic inference (e.g., the semantics of expressions), and as “low-level”, those we take to constrain the specific task (e.g., cognitively or physically) in which the linguistic inferences of interest are hypothesized to be implicated.

²See Jasbi, Waldon, and Degen 2019 and Waldon and Degen 2020 for recent approaches to modeling linking hypotheses in formal inference and truth value judgment experiments.

challenging, since inferences are greatly influenced by a predicate’s context of use (Karttunen 1971 *et seq.*). To better understand the factors driving factive inferences, many researchers have begun to collect native speaker judgments in formal experiments, using them to evaluate hypotheses about the semantic properties of potentially factive predicates, as well as hypotheses about how these properties are reflected in the distributional patterns observed in judgment data (Tonhauser 2016; Djärv and Bacovcin 2017; Djärv, Zehr, and Schwarz 2018; White and Rawlins 2018b; White, Rudinger, et al. 2018; White 2021; Degen and Tonhauser 2021; Degen and Tonhauser 2022; Jeong 2021; Kane, Gantt, and White 2022).

Of particular concern in the experimental literature on factivity has been the observation that, in tasks aimed at measuring a predicate’s factivity, aggregate measures of inference judgments show much more gradience than one might expect under the classical view of factivity as a discrete lexical property (White and Rawlins 2018b). Some authors have gone so far as to claim that the observed gradience indeed casts doubt on this view (Degen and Tonhauser 2022, building on Simons 2001 *et seq.*). Doubt about the existence of such a lexical property is furthermore consistent with the view that projection, in general, is fundamentally gradient in nature (Tonhauser, Beaver, and Degen 2018). This *Fundamental Gradience Hypothesis* contrasts with a *Fundamental Discreteness Hypothesis*, under which projection—and specifically, factivity—comprises a discrete property or collection of properties of an expression on a particular occasion of use. We discuss these hypotheses in more detail in Section 2.

Our central aim in this paper is to quantitatively evaluate these hypotheses in terms of a framework that allows us to explicitly formulate the link between their respective construals of factivity and the way humans produce inference judgments. The core theoretical contribution we make in employing this framework, which is developed in Grove and White 2025b, is to transparently relate the sorts of compositional analyses of the meanings of factive predicates common in the formal semantics literature to probabilistic models characterizing distributions over inference judgments.

We outline the main ingredients of this framework in Section 3 before using it to carry out a comparison between the Fundamental Gradience Hypothesis and the Fundamental Discreteness Hypothesis in Sections 4 and 5 (see Grove and White 2025b for an introduction to the framework in its full generality). As we show, the framework of Grove and White allows us to precisely target where the two hypotheses make different predictions about the distribution of inference judgments across participants, making an apples-to-apples comparison feasible. Moreover, it allows us to do this using standard statistical model comparison metrics that balance out a model’s fit to inference judgment data against the model’s complexity.

Using such metrics, we find that models that implement the Fundamental Discreteness Hypothesis unambiguously outperform models that implement the Fundamental Gradience Hypothesis, across both an existing dataset aimed at measuring projective inferences (Degen and Tonhauser 2021) and three new datasets: a replication of the existing dataset, as well as two new datasets that modify Degen and Tonhauser’s paradigm to evaluate model robustness. In Section 6, we argue that these results favor an account of factivity that regards it as fundamentally discrete in nature, and we discuss how this discreteness might be understood within classical semantic accounts

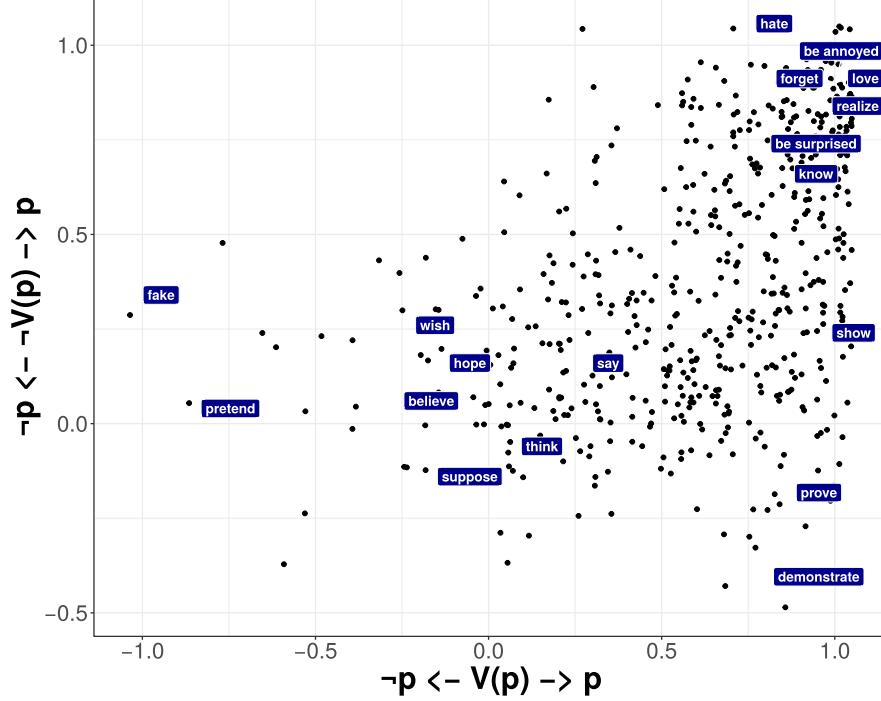


Fig. 1 An aggregate measure of factivity, derived from the MegaVeridicality dataset of White and Rawlins 2018b. Points are jittered (0.05) to avoid overplotting. The x -axis corresponds to the mean scores, across participants, of responses to prompts of the form *Someone (was) ...ed that a particular thing happened. Did that thing happen?* with possible responses *yes* (mapped to 1), *maybe or maybe not* (mapped to 0), and *no* (mapped to -1). The y -axis corresponds to the same measure for prompts of the form *Someone {didn't, wasn't} ... that a particular thing happened. Did that thing happen?* with the same possible responses. Each point is a predicate, with a selection of points labeled. Note that jittering sends some of the points above or below 1 and -1, respectively.

of factive predicates, as well as accounts that attempt to reduce presupposition projection to entirely pragmatic processes (Simons 2007; Simons, Tonhauser, et al. 2010; Simons, Beaver, et al. 2017).

2 Gradient inference patterns among factive predicates

Inference judgment datasets—such as MegaVeridicality (White and Rawlins 2018b; White, Rudinger, et al. 2018), VerbVeridicality (Ross and Pavlick 2019), and CommitmentBank (De Marneffe, Simons, and Tonhauser 2019)—have enabled fine-grained analyses of inference judgment patterns across the entire clause-embedding lexicon. Remarkably, the aggregate judgments of multiple speakers seen in such datasets reveal substantial gradience, a fact which has garnered sustained focus in recent work.

In the domain of factivity, White and Rawlins (2018b) observe gradience in the aggregate measures of various predicates' degrees of factivity, using data from the

MegaVeridicality dataset (summarized in Figure 1; see also Xue and Onea 2011; Smith and Hall 2011; Djärv and Bacovcin 2017 for related observations). Crucially, they note the lack of clear divisions among classes of predicates and suggest that speakers’ gradient aggregate veridicality ratings are likely influenced by the fine-grained semantics of particular verbs (*ibid.*, p. 228; see Roberts and Simons 2024 for recent discussion).³

In later work, Degen and Tonhauser (2022) investigate the nature of this gradience in six experiments and argue that its persistence rebuts the hypothesis that there is a coherent class of factive predicates. Our own modeling work uses data collected under the same experimental paradigm as that which Degen and Tonhauser employ, so we describe their data and arguments in detail in Section 2.1. In Section 2.2, we turn to the broad question of which factors may be responsible for the gradience observed in inference judgment tasks. We outline an array of explanatory choices that one can make in attempting to account for this gradience, and we discuss some of the evidence that would be pertinent to these choices. We then present two alternative hypotheses about the nature of gradience among inference judgments that constitute the main focus of this paper—the Fundamental Discreteness Hypothesis and the Fundamental Gradience Hypothesis—and we relate them to the kinds of explanatory choices we have introduced. In Section 2.4, we discuss the specific paradigm we use to evaluate these two hypotheses.

2.1 Measuring veridicality and factivity

In all of their experiments, Degen and Tonhauser (2022) focus on the set of twenty clause-embedding predicates listed in (3), which they group into hypothetical classes that follow the prior literature on factivity (Kiparsky and Kiparsky 1970; Karttunen 1971; Hooper and Thompson 1973; Givón 1973; Hooper 1975; Abusch 2002; Abusch 2010; Abrusán 2011; Abrusán 2016; Anand and Hacquard 2014, i.a.).

- (3) Twenty clause-embedding predicates (Degen and Tonhauser 2022, p. 559, ex. 13)
 - a. canonically factive: *be annoyed, discover, know, reveal, see*
 - b. non-factive
 - (i) non-veridical non-factive: *pretend, say, suggest, think*
 - (ii) veridical non-factive: *be right, demonstrate*
 - c. optionally factive: *acknowledge, admit, announce, confess, confirm, establish, hear, inform, prove*

In their discussion of the experimental data which they go on to collect, they suggest that insofar as the standard classification of predicates is correct, “...we expect to see a categorical difference in projection between canonically factive predicates on the one hand, and optionally factive and nonfactive predicates on the other” (p. 569). To measure projection, Degen and Tonhauser provide participants with a description of a scenario in which someone asks a polar question whose main verb is one of the clause-embedding predicates of interest, e.g., (4).

³White and Rawlins are primarily interested in the relationship between the veridicality inferences associated with particular predicates and those predicates’ syntactic distributions—not their semantic classifications.

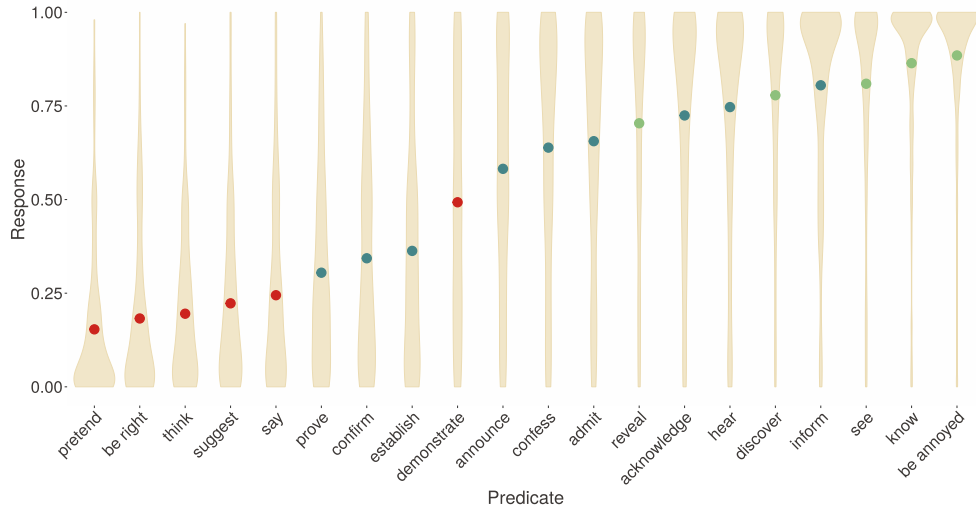


Fig. 2 Verb means from Degen and Tonhauser’s (2022) experiment 1a. “Non-factive” verbs are in red, “optionally factive” verbs are in teal, and “canonically factive” verbs are in green. Violin plots indicate the probability density of responses. (See Degen and Tonhauser 2022, Figure 2, p. 562.)

(4) **Helen asks:** Did Amanda discover that Danny ate the last cupcake?

They then ask participants to provide a rating on a continuous slider scale from *no* to *yes* in answer to a prompt of the form in (5). This prompt is meant to assess the extent to which participants believe that the content of the embedded clause projects, using speaker certainty as a proxy for projection.

(5) Is Helen certain that Danny ate the last cupcake?

In another experiment, they give participants a variant of this task in which their answer is provided as a binary forced choice between *no* and *yes*, rather than as a slider scale response.

Consistent with White and Rawlins’s original observations, Degen and Tonhauser find that the patterns of inference ratings across predicates in their experiments are gradient in nature for both projection and veridicality inferences. Indeed, the degree to which predicates display projective inferences appears to evolve continuously from the least projective predicate (*pretend*) to the most projective predicate (*be annoyed*) when predicates are compared in terms of their mean ratings (see Figure 2). This gradience is manifest in both the experiment which collects sliding scale judgments and the one which collects binary judgments.⁴

2.2 Gradience in inference datasets

Degen and Tonhauser’s results are consistent not only with White and Rawlins’s original observations, but also with findings in adjacent domains. An and White (2020)

⁴In addition to measuring projection, Degen and Tonhauser measure *veridicality* inferences in two other experiments and find a similar pattern. We omit a discussion of these experiments for reasons of space.

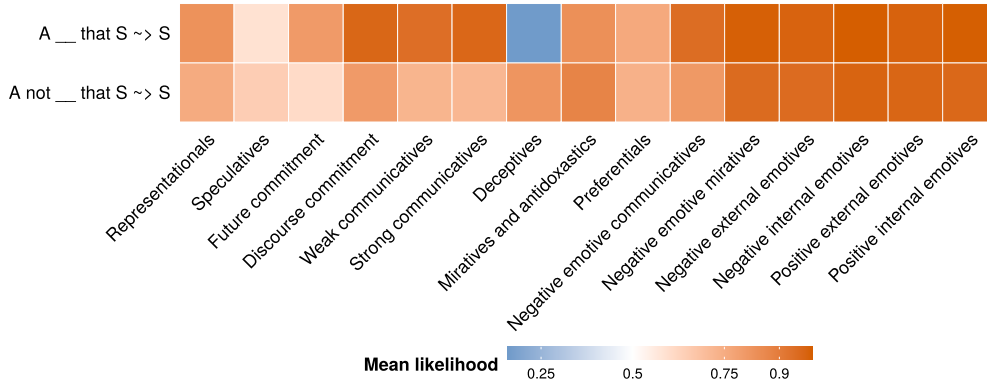


Fig. 3 Veridicality inferences associated with each class found in Kane, Gantt, and White 2022. Kane, Gantt, and White provide labels for each class based on the predicates that occur in that class as well as the belief and desire inferences associated with that class. The top row and bottom row correspond to the x - and y -axes of Figure 1, respectively. The classes associated with dark orange cells in both rows are taken to be factive subclasses.

observe similar gradience in neg-raising inferences, as captured in their MegaNegRaising dataset; and Kane, Gantt, and White (2022) note a similar pattern among the belief and desire inferences they obtain in their MegaIntensionality dataset.

Kane, Gantt, and White note that in the face of such gradience, it is reasonable to entertain two views. On the one hand, it is possible that “apparent gradience indicates that no formally represented lexical property controls whether a particular inference is triggered” (p. 572). On the other hand, “apparent gradience [may be] partly or wholly a product of the methods often used to collect inference judgments, and [it may be] that there are discrete, formally represented lexical properties that are active in triggering... inferences” (p. 572). To pursue this question, they ask whether clear *patterns* emerge across the inference judgment datasets discussed above—MegaVeridicality, MegaNegRaising, and MegaIntensionality—by clustering predicates into classes sensitive to the responses from those datasets so as to optimize their ability to predict predicates’ syntactic distributions, as measured in the MegaAcceptability dataset (White and Rawlins 2016).⁵ They uncover fifteen classes that correspond extremely closely to those that one would expect from prior work on clause-embedding predicates. As shown in Figure 3, these classes include a variety of factive subclasses that differ principally in the patterns of belief and desire inferences they are associated with. As one might expect (Karttunen 1971 *et seq.*), the true factive subclasses tend to be emotive and include, for example, *love* and *hate*.

Thus Kane, Gantt, and White’s findings establish that there *is* a coherent class of factive predicates—the *emotives*—which are, in turn, subclassed by the belief and desire inferences they give rise to. But they also find that there are a variety of classes associated with weaker veridicality inferences than one might expect from a truly factive class. These classes include non-emotive predicates, like *know* and *realize*. Thus

⁵The idea behind optimizing the predictability of predicates’ syntactic distributions is that insofar as the classes to which a predicate belongs are predictive of the predicate’s syntactic distribution, there is preliminary evidence that such classes are associated with distinct distributionally active lexical representations.

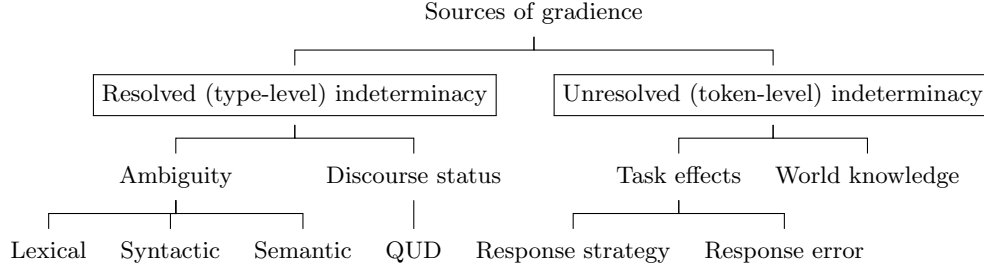


Fig. 4 A partial taxonomy of the sources of gradience in inference judgment tasks.

while it is not correct to say that there is *no* class of factive predicates, one must still explain the apparent gradience associated with certain classes, such as the non- emotive ones, as Degen and Tonhauser point out (their Section 4.1, Objection 3). Class-level gradience of this kind is unlikely to be—as Kane, Gantt, and White put it—“partly or wholly a product of the methods often used to collect inference judgments”, since their analysis expressly accounts for the relevant task effects.

2.3 Sources of gradience

How, then, should one account for gradience, whether it is due to knowledge about particular predicates or the classes those predicates fall into? To approach this question, we regiment the possible sources of gradience into two broad classes: *resolved* (or *type-level*) indeterminacy and *unresolved* (or *token-level*) indeterminacy. A partial taxonomy of these sources is provided in Figure 4.⁶

2.3.1 Resolved indeterminacy

Resolved indeterminacy drives any uncertainty about the *type* of speech act one is drawing an inference from. Consider an utterance of the sentence *my uncle is running the race*, where the verb *run* is ambiguous: the speaker’s uncle might be a participant in the race, or he might be in charge of its logistics. If one is presented with this sentence and asked, “On a scale of 1 to 10, how likely is it that my uncle has good managerial skills?”, one might be tempted to pick a lower number—say, around 2—if *run* is interpreted in its locomotive sense, while one might be tempted to pick a higher number—say, around 8—if *run* is interpreted in its managerial sense. If a hundred people are asked the same question, perhaps about half of them will interpret *run* in the locomotive sense, while the other half interpret it in the managerial sense. And while the distribution of judgments will look bimodal—with one mode around 2 and the other mode around 8—the *average* judgment will be around 5. Gradience that stems from a semantic ambiguity is thus observable at the level of an entire experiment

⁶This taxonomy is only partial (and admittedly, based on conjecture). For instance, Chris Kennedy (p.c.) points out that expressions displaying *open texture*—the pattern of uncertainty about the extensions of, e.g., non-gradable predicates such as *sandwich* (is a hot-dog a sandwich?)—seem likely to give rise to gradience (Mackinnon, Waismann, and Kneale 1945), but it is potentially contentious whether open texture should be thought of as type- or token-level indeterminacy.

(i.e., an inference judgment dataset), rather than at the level of an individual trial (a single observation in that dataset).

Other likely examples of resolved indeterminacy involve ambiguities relating to, e.g., quantifier scope and binding, or (possibly more controversially) discourse-level properties, such as the question under discussion (QUD; Ginzburg 1996; Roberts 2012). Each may lead to distinct all-or-nothing inferences produced on different utterance occasions.

2.3.2 Unresolved indeterminacy

Unresolved indeterminacy drives uncertainty that persists about the status of an inference even after having fixed the meanings of the expressions used to perform a particular speech act. One kind of unresolved indeterminacy may be *vagueness*: from an utterance of *my uncle is tall*, there can be uncertainty about whether or not my uncle is at least six feet tall. Another comes from *world knowledge*: given an utterance of *my uncle is running the race* (and having fixed *run* to its locomotive sense), there can still be uncertainty about how fast he runs, given that he is a runner.

Crucially, such indeterminacy would be reflected on *individual trials* of an inference judgment experiment. Moreover, resolved indeterminacy and unresolved indeterminacy can coexist: in the hypothetical inference judgment task described above, there is uncertainty about the intended sense of the verb *run*, and there is uncertainty about the inference that the referent of *my uncle* has good managerial skills, which itself stems from world knowledge.

Finally, unresolved indeterminacy in an experimental setting might also stem from task effects, such as one’s physical response error or one’s response strategy; for example, some experimental participants might hedge their responses so that they are never quite at the extremes of the relevant measurement scale. This kind of indeterminacy persists even once the interpretation of an expression is fixed.

2.3.3 The application to factivity

In the realm of factive inferences, one could adopt the view that gradience among individual predicates and classes of predicates stems from unresolved indeterminacy. This view would align with Tonhauser, Beaver, and Degen’s *Gradient Projection Principle* (ex. 7, p. 499):

- (6) **Gradient Projection Principle:** If content C is expressed by a constituent embedded under an entailment-canceling operator, then C projects to the extent that it is not at-issue.

Alternatively, one might adopt the view that such indeterminacy is actually resolved. As Degen and Tonhauser put it, the latter view says that “the observed gradience in projection [is] compatible with a binary factivity category in combination with two assumptions: first, that predicates may be ambiguous between a factive lexical entry . . . and a nonfactive lexical entry . . . and, second, that interpreters may be uncertain about which lexical entry a speaker intended in their utterance” (p. 583). Our task in this

paper is to formalize these two types of explanation so that we may quantitatively compare them.

We consider two hypotheses about the sources of gradience in projective inference judgments among clause-embedding predicates:

- (7) a. *The Fundamental Discreteness Hypothesis.* Factivity is a discrete property of at least some token occurrences of at least some clause-embedding predicates (or expressions containing them). A given use of (an expression containing) a clause-embedding predicate either triggers a projective inference, or it does not.
- b. *The Fundamental Gradience Hypothesis.* There is no property distinguishing factive from non-factive occurrences of clause-embedding predicates. Instead, the gradient distinctions observed among predicates and their classes reflect the gradient contributions different predicates make to inferences about the truth of their complement clauses.

According to the Fundamental Discreteness Hypothesis, the gradience associated with the predicates in (3) is due to resolved indeterminacy. Different occasions on which a predicate is used may vary with respect to whether or not the predicate triggers a projective inference, but such variation concerns which of the alternative interpretations of the predicate should be selected. Alternative interpretations may be available for a variety of reasons: among other possibilities, (i) a predicate may have multiple senses—at least one that is implicated in triggering projection and at least one that is not; (ii) a predicate may occur in multiple structures—at least one that is implicated in triggering projection and at least one that is not;⁷ or (iii) a predicate may be usable in contexts where the common ground—or some other construct, such as the question under discussion—entails the content of its embedded clause, as well as contexts where it does not (see Simons, Beaver, et al. 2017; Roberts and Simons 2024).⁸ The gradience associated with particular classes of predicates that Kane, Gantt, and White observe might then indicate (respectively) (i) that there is regular polysemy among predicates within a class; (ii) that predicates within a class bias the resolution of syntactic ambiguity in similar ways; or (iii) that predicates within a class tend to show up in similarly structured contexts. It is noteworthy that possibility (iii) is consistent with the *Projection Principle* of Beaver et al. (2017, ex. 33, p. 280), which one might interpret as a *discrete* variant of the Gradient Projection Principle:

- (8) **Projection Principle:** If content C is expressed by a constituent embedded under an entailment-canceling operator, then C projects if and only if C is not at-issue.

On this view, certain (classes of) predicates may tend to show up in contexts in which the contents of their complements are (or aren't) at-issue. We stress that the

⁷The second option is plausible in light of cross-linguistic evidence that functional items surrounding a predicate can modulate veridicality inferences; e.g., nominal morphology attached to verbs or their clausal complements (see Varlokosta 1994; Giannakidou 1998; Giannakidou 1999; Giannakidou 2009; Roussou 2010; Farudi 2007; Abrusán 2011; Kastner 2015; Ozyildiz 2017; see White 2019 for discussion).

⁸See Qing, Goodman, and Lassiter 2016 for an approach within the Rational Speech-Act framework (Frank and Goodman 2012; Goodman and Stuhlmüller 2013) along the lines of possibility (iii).

Fundamental Discreteness Hypothesis does not dictate whether the indeterminacy that drives the observed gradience is relevant to any particular semantic or pragmatic construct—only how that indeterminacy is resolved.

According to the Fundamental Gradience Hypothesis, the observed gradience is a product of unresolved indeterminacy: it is more like vagueness than it is like ambiguity. This hypothesis is therefore consistent with Tonhauser, Beaver, and Degen’s (2018) Gradient Projection Principle. Tonhauser, Beaver, and Degen themselves consider two interpretations of this principle. On the one hand, they suggest that “a listener’s (or reader’s) judgment that a content is projective to a certain extent means that the listener takes the speaker (or writer) to be committed to the content to that extent” (p. 498); and on the other, they consider that “a listener’s judgment that a content is projective to a certain extent reflects the probability with which they believe the speaker to be committed to the content” (pp. 498–499).

The first of these interpretations is unambiguously consistent with Fundamental Gradience. The second interpretation has a more ambiguous status and might, in principle, be seen as consistent with either of the hypotheses. As Tonhauser, Beaver, and Degen say, “[o]n this interpretation, speaker commitment may be a binary, categorical property and projection variability arises from the listener’s uncertainty about the whether the speaker is committed” (p. 499). That is, Tonhauser, Beaver, and Degen consider the possibility that a speaker’s commitment might be discrete, while the uncertainty that informs a listener’s judgment might be gradient. This possibility might be viewed as consistent with the Fundamental Discreteness Hypothesis insofar as it allows that listeners end up making a judgment about whether they take the speaker to be committed to the content of the embedded clause in spite of their uncertainty, rather than retaining that uncertainty as the discourse develops.

For the purposes of this paper, we interpret the Gradient Projection Principle to imply the view that gradience in patterns of projection arises from unresolved indeterminacy. We believe this interpretation is reasonable in light of the fact that the alternative, under which the Gradient Projection Principle is consistent with the Fundamental Discreteness Hypothesis, reduces it to the vanilla Projection Principle.

2.4 The role of world knowledge in gradient inference judgments

Our tests of these hypotheses will rely on a paradigm used by Degen and Tonhauser (2021), who aim to characterize the influence of world knowledge on projective inferences, focusing on the same twenty clause-embedding predicates as in (3). We employ this paradigm because it allows us to explicitly model the influence of non-semantic factors—specifically, speakers’ prior beliefs about the likelihood of an inference being true. Similar to the experiment reported above, in which Degen and Tonhauser (2022) measure projection out of the complement of a predicate placed inside of a polar question, Degen and Tonhauser (2021) measure projective inferences in the presence of a background fact whose content they manipulate (their experiment 2b). For example, the following experimental trial features the predicate *pretend*:

Fact (which Elizabeth knows): Zoe is a math major.

Elizabeth asks: "Did Tim pretend that Zoe calculated the tip?"

Is Elizabeth certain that Zoe calculated the tip?

no

yes

Next

The same twenty complement clauses as featured in Degen and Tonhauser 2022 are also featured in this experiment, but each clause is now paired with one of two background facts, constructed so that one fact makes the clause more likely to be true than the other one does. Each participant in this experiment sees twenty items (along with six control items). On each experimental trial, a predicate is placed in the context of one of the twenty clauses, along with one of the two background facts constructed for that clause. The results Degen and Tonhauser (2021) obtain in this setting mirror those of Degen and Tonhauser (2022): the mean projection ratings for the twenty predicates show a similar gradient pattern (Spearman's $r = 0.98$).

In addition to the assessment of projective inferences given background facts, Degen and Tonhauser (2021) conduct a norming experiment, in which the prior certainties about the truth of the complement clauses featured in their projection experiment are assessed independently, given the same background facts (their experiment 2a). Trials in this experiment ask participants to judge how likely the relevant clause is to be true, given one of the two background facts constructed for it. For example, the following trial features the same clause as in the example provided previously, but with the alternate low-probability fact:

Fact: Zoe is 5 years old.

How likely is it that Zoe calculated the tip?

impossible

definitely

Continue

Degen and Tonhauser find that the by-item means for the forty pairs of complement clauses and background facts, as assessed in their norming experiment, are a good linear predictor of the inference ratings for items featuring the same complement clauses and facts which they obtain in their experiment investigating projective inferences.

Hence, at least one source of the variation among the projective inferences associated with clause-embedding predicates is the *context* in which these predicates are placed; and in particular, the prior certainties that people associate with these contexts. This cannot be the whole story, of course, as Degen and Tonhauser observe: the mean projection ratings for predicates display substantial gradience even after collapsing across the contexts in which they occur (consistent with their 2022 experiment 1a: see Figure 2). So, what explains the remaining variation?

In the next section, we introduce a framework that allows us to precisely state *theories* of gradience and to rigorously relate these theories to probabilistic models of inference judgment data. When constructing such theories, we attempt to remain relatively abstract, and to include only as much semantic detail as we need in order to distinguish the two hypotheses in (7). At the same time, we aim to be clear about how one could elaborate on our account, as implemented, in order to spell out more fine-grained variants of the hypotheses.

3 Probabilistic semantics

To state a theory of gradience precisely, it is useful to have a general method of integrating probabilistic reasoning into a compositional semantics of English. Here we rely on the framework of Probabilistic Dynamic Semantics (PDS; Grove and White 2025b), which follows earlier work by Grove and Bernardy (2023) in adopting an interface for performing Bayesian reasoning in typed λ -calculus using *monads*.⁹ Our reason for employing this framework is that it allows one to transparently relate the sorts of compositional analyses of expressions’ meanings common in formal semantics to probabilistic models that characterize distributions over inference judgments.

Importantly, the framework allows the two hypotheses laid out above to be precisely specified, while keeping fixed both the formal analysis of the expressions of interest and the way in which probability distributions over inference judgments are mapped onto a particular data collection instrument. These aspects of the framework are important because (as we show in Sections 4 and 5) they allow us to conduct an apples-to-apples comparison of the two hypotheses which both (a) precisely targets where they make different predictions about the distribution of inference judgments across experimental trials, and (b) does so using standard statistical model comparison metrics which balance a model’s fit to inference judgment data against its complexity.

We begin in Sections 3.1 to 3.5 with relevant background on Grove and White’s framework before turning in Section 3.6 to an example. Our aim is to show how one may finely delineate indeterminacy that is core to the semantic value of an expression—giving rise to unresolved indeterminacy—from indeterminacy about which interpretation should be associated with a particular string—giving rise to resolved

⁹See Giorgolo and Asudeh 2014; Asudeh and Giorgolo 2020 and Bernardy et al. 2018 for related monadic approaches. The aforementioned works have somewhat different aims from Grove and Bernardy’s, which are reflected in the distinct interfaces they supply.

indeterminacy (see also Bergen, Levy, and Goodman 2016; Potts et al. 2016; Monroe 2018 for related approaches). To illustrate the distinction between these forms of indeterminacy, we provide an example involving gradable adjectives. Then we provide an analysis of factivity in Section 3.7.

3.1 Distributions on discourse constructs

Before launching into a probabilistic analysis of clause-embedding predicates, we lay out the core principles of PDS—in particular, how it supports dynamic analyses of discourse phenomena and the lexical semantics of expressions which trigger them, how it incorporates probabilistic knowledge into these analyses, and how it may be used to generate predictions about people’s behavior in the context of particular linguistic inference tasks.

In PDS, an ongoing discourse is represented by a map from an input state to a probability distribution over output states. States are tuples of *metalinguistic parameters*; these include, e.g., the common ground (or *context set*; Stalnaker 1978) and the QUD, as well as other conversationally relevant features of discourse (e.g., possible antecedents for elliptical constructions). One can view the state as akin to the context state of Farkas and Bruce (2010), which includes representations such as a QUD stack (Roberts 2012) and a common ground, among other constructs; PDS states are less constrained than Farkas and Bruce states, in principle, in the sense that they may define arbitrary discourse constructs.

We notate the types of these state tuples ‘ σ ’, ‘ σ' ’, etc., so that an ongoing discourse is of type $\sigma \rightarrow P\sigma'$, where $P\sigma'$ is the type of *probability distributions* over states of type σ' . In words: a discourse is a map from input states to probability distributions over output states. Generally, $P\alpha$ is the type of probability distributions over values of type α . When we have a term of type $P\alpha$, we sometimes refer to it as a *probabilistic program*.

What is a probabilistic program? In PDS, $P\alpha$ is a type inhabited by λ -terms, just like all other types are. But whereas a λ -term that inhabits a function type, $\alpha \rightarrow \beta$, is a function which, given something of type α , computes something of type β , a λ -term that inhabits the type $P\alpha$ is a computation of a random value of type α according to some probability distribution. How exactly this computation ends up being carried out—whether, e.g., by analytically computing a closed-form representation of the density associated with an arbitrary value of type α , or by approximating this density by sampling values directly from the relevant distribution, or by yet some other method—is immaterial. The purely declarative language of λ -terms makes space to do one thing only: describe the *sort* of computation of type $P\alpha$ which must be carried out in order to compute a random value of type α . Describing such a computation answers questions such as, “What information is this random value conditional on?”, “Does it depend somehow on a value computed from a particular distribution?”, “Are there values it is blocked from taking on?”. That is, it provides a complete mathematical description of any random variable of interest in terms of the dependencies it enters into in virtue of the control flow of the program. By remaining indifferent about implementation details, we are afforded a useful abstraction that puts probabilistic programs on the same plane as other typed λ -terms. This abstraction allows such

programs to be integrated into the ambient calculus seamlessly, without changing its meaning. The monadic interface endows probabilistic programs with the technology required to compose them together while preserving their internal structure.

To perform updates of various types to an ongoing discourse, we use two operators—*bind* and *return*—which allow probability distributions to be sampled (bind) and ordinary, non-probabilistic values to be returned as *degenerate* distributions over those values (return).¹⁰ Both bind and return are indispensable if one’s aim is to characterize arbitrary probability distributions *compositionally*—i.e., to construct programs any of whose syntactic parts has a fixed (e.g., model-theoretic) interpretation. An interface for characterizing programs compositionally is indeed *necessary* if we also wish to characterize natural language meanings compositionally, where such meanings *depend* on probabilistic programs. If we want to compose linguistic meanings using functional application (for example), then our compositional regime already has its own interface with an attendant structure: we must view both the function and the argument—which may depend on a probability distribution—as objects that inhabit the right kind of space. The monadic treatment renders both interfaces fully compatible, allowing us to speak about functional application and probabilistic uncertainty in the same breath. As a result, we can develop semantic theories of natural language without compromising on either the semantic interface or the probabilistic interface at any point.

To illustrate, suppose we have some categorical distribution $\mathbf{mammal} : \mathbf{Pe}$ on mammals (where e is the type of entities). We can represent a distribution on mammals’ mothers as:

$$\begin{array}{c} x \sim \mathbf{mammal} \\ \boxed{\mathbf{mother}(x)} \end{array}$$

Here, $\mathbf{mother} : e \rightarrow e$ maps an entity to its mother. x is sampled from $\mathbf{mammal}$ using bind, and $\mathbf{mother}(x)$ is then *returned*, as notated by the box. The idea is that the resulting distribution assigns to any given entity y the probability $\sum_{x \in \mathbf{mother}^{-1}(y)} p_{\mathbf{mammal}}(x)$ (where $\mathbf{mother}^{-1} : e \rightarrow e \rightarrow t$ is the inverse of the $\mathbf{mother}$ function and $p_{\mathbf{mammal}}(x)$ is the probability that $\mathbf{mammal}$ assigns to x).

In a similar fashion, given two discourses $D_1 : \sigma \rightarrow \mathbf{P}\sigma'$ and $D_2 : \sigma' \rightarrow \mathbf{P}\sigma''$, we can invoke bind to sequence D_1 with D_2 and obtain a new discourse $D_3 : \sigma \rightarrow \mathbf{P}\sigma''$:

$$D_3 = \lambda s. \left(\begin{array}{c} s' \sim D_1(s) \\ D_2(s') \end{array} \right)$$

To use a “sampling” metaphor: given a starting state s , we sample an output state from $D_1(s)$ and apply D_2 to this output state. Further, given an input state s , the probability $p_{D_3(s)}(s'')$ that $D_3(s)$ assigns to an output state s'' is computed as:

$$p_{D_3(s)}(s'') = \sum_{s'} p_{D_2(s')}(s'') * p_{D_1(s)}(s')$$

¹⁰Importantly, these operators conform to the monad laws. See Grove and Bernardy 2023; Grove and White 2025b for why this fact is useful.

In words, sequencing two maps from input states to probability distributions over output states involves doing pointwise multiplication of the probabilities assigned to intermediate states by the first map and those assigned to an output state by the second map and then summing the result.

Finally, we represent the common ground as a probability distribution over indices representing information about possible worlds, along with certain linguistic parameters (which, for current purposes, we keep fairly open ended); we refer to the type of indices as ι , so that common grounds themselves are distributions of type $P\iota$. Given some index $i : \iota$, we refer to the “world” of i as w_i and to what we call the “context” of i as c_i ; contexts are what we use to encode useful linguistic information. The idea is that worlds w determine, e.g., how tall a particular individual is, while contexts c determine, e.g., the vague threshold of height past which one’s height is considered tall. Thus while a world-sensitive function such as $\text{height}(i) : e \rightarrow r$ (which we sometimes write $\text{height}(w_i)$) intuitively says something about how tall an entity is at an index i , the adjective’s threshold $\text{d}_{\text{tall}}(i) : r$ (which we sometimes write $\text{d}_{\text{tall}}(c_i)$) only provides information about an aspect of the lexical semantics of *tall* at that index, viz., what is required of an entity to count as tall.

3.2 Making an assertion

Similar to discourses, the meanings of expressions are both probabilistic and dynamic. Accordingly, we represent them as functions of type $\sigma \rightarrow P(\alpha \times \sigma')$: given an input state, the meaning of an expression produces a probability distribution over *pairs* of ordinary meanings of type α and possible output states. Given a sentence whose probabilistic dynamic meaning ϕ is of type $\sigma \rightarrow P((\iota \rightarrow t) \times \sigma')$, we can represent an *assertion* of that sentence as a discourse which updates the common ground. Specifically, we have a function **assert** with the following type:

$$\text{assert} : (\sigma \rightarrow P((\iota \rightarrow t) \times \sigma')) \rightarrow \sigma \rightarrow P\sigma'$$

Given such a ϕ , **assert**(ϕ) is a discourse of type $\sigma \rightarrow P\sigma'$ representing an assertion of ϕ . For any input state $s : \sigma$, **assert**(ϕ)(s) samples a proposition p together with an output state s' from ϕ and updates the common ground of s' with p ; in particular, it gives back a new state s'' just like s' , except that the common ground of s'' ($\text{CG}(s'')$) is just like the common ground of s' ($\text{CG}(s')$), but updated with p . Ultimately, assertions modify an ongoing discourse so that its probability distribution over output states features common grounds in which the proposition returned by ϕ has been observed to hold true (see Grove and White 2025b for further details).

3.3 Asking a question

We follow a categorial tradition by analyzing questions as denoting—given an index—sets of true short answer meanings (Hausser and Zaefferer 1978; Hausser 1983; Xiang 2021; cf. Karttunen 1977; Groenendijk and Stokhof 1984). For simplicity, we analyze only degree questions: given an index, these denote sets of degrees of type r (real numbers). Questions therefore have probabilistic dynamic meanings of type $\sigma \rightarrow P((r \rightarrow \iota \rightarrow t) \times \sigma')$.

Asking a question is a matter of pushing a question meaning onto the QUD stack (Roberts 2012; Farkas and Bruce 2010). Reflecting this, there is a function `ask` with the following type:

$$\text{ask} : (\sigma \rightarrow \mathbf{P}((r \rightarrow \iota \rightarrow t) \times \sigma')) \rightarrow \sigma \rightarrow \mathbf{P}(\text{QUD}\sigma')$$

Here, $\text{QUD}\sigma'$ is the type of a new state, formed out of a state of type σ' by adding a degree question to the top of its QUD stack. Given a probabilistic dynamic question meaning κ and an input state s , $\text{ask}(\kappa)(s)$ samples a pair of a question meaning $q : r \rightarrow \iota \rightarrow t$ and a state $s' : \sigma'$ from $\kappa(s)$ and then updates the QUD stack of s' by pushing q onto it.

3.4 Responding to a question

At any point in an ongoing discourse in which the QUD stack is not empty, one can respond to the QUD at the top of the stack based on one's prior knowledge. Concretely, a given responder has some background knowledge $bg : \mathbf{P}\sigma$ constituting a prior distribution over *starting* states. The responder uses this prior, in conjunction with the interim updates to the discourse, to derive a probability distribution over answers to the QUD. This answer distribution is gotten by retrieving the QUD from the state and choosing a degree of which this QUD is true, given an index sampled from the common ground. The result is a distribution over degree answers.

3.5 Linking assumptions

In the setting of a formal experiment, an answer needs to be given using a particular testing instrument. We model such instruments using a family f of distributions representing the likelihood, which is then fixed by a collection Φ of nuisance parameters. We thereby define a family of *response functions*, parametric in the testing instrument (i.e., likelihood function), each of which takes two arguments—a distribution representing one's background knowledge and an ongoing discourse:

$$\text{respond}^{f_\Phi : r \rightarrow \mathbf{P}\rho} : \mathbf{P}\sigma \rightarrow (\sigma \rightarrow \mathbf{P}(\text{QUD}\sigma')) \rightarrow \mathbf{P}\rho$$

For a fixed likelihood f_Φ mapping degree answers onto a distribution over possible responses of type ρ (for some ρ), the response function produces a response distribution. It does this by composing the discourse with the background knowledge distribution, via an application of `bind`, and then obtaining a degree answer to the current QUD, before applying the likelihood function f_Φ to this degree (see Section 4.1.1 for further details).

The testing instrument we use in our experiments is a slider scale that records responses on the unit interval $[0, 1]$. One suitable likelihood is therefore a *truncated normal distribution* with standard deviation σ : $f(x, \Phi) = \mathcal{N}(x, \sigma) \mathbf{T}[0, 1]$ (so that $f = \mathcal{N}$ and $\Phi = \sigma$). This likelihood—which we employ in our models of factivity—can be viewed as allowing some distribution of response errors, given the intended target response, i.e., the answer to the question. Crucially, it stays within the bounds $([0, 1])$ of the slider scale.

3.6 Two kinds of uncertainty

We rely on the distinction encoded in PDS between uncertainty about the discourse state and uncertainty in the common ground to model the general behavior of resolved and unresolved indeterminacy, respectively. For illustration, take the sentence in (9).

(9) My hands are dry.

As Kennedy and McNally (2005) note, (9) contains an ambiguity brought on by the two senses of the adjective *dry*: on one reading, (9) is a claim about the quality of the speaker’s skin, while on another, it is a claim about the amount of liquid with which their skin is in contact. Kennedy and McNally argue that these two senses of *dry* associate it with *relative* (i.e., vague) and *absolute* standards of dryness, respectively. While relative standards are theorized to shift from one context of use of the adjective to another, absolute standards are theorized (a) to be lexically fixed and, generally, (b) to coincide with an *endpoint* on the scale that the adjective lexically encodes. Kennedy and McNally show that the two relevant senses can be brought out by modifiers which are typically compatible with only one or the other type of standard.

- (10) a. My hands are very dry.
b. My hands are partially dry.

While (10a) has the “skin quality” sense (the relative sense), (10b) has the “contact with liquid” sense (the absolute sense).

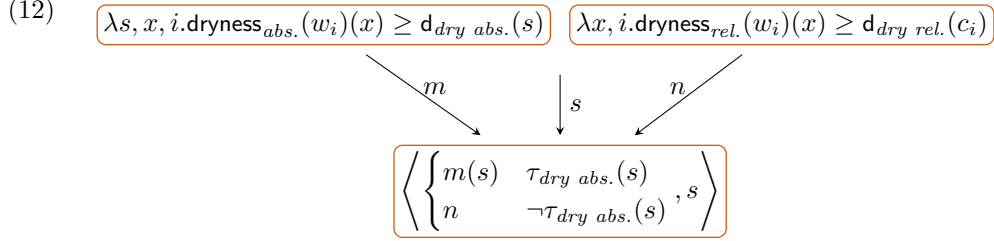
We can model this ambiguity by interpreting *dry* as the probabilistic semantic value in (11)—something of type $\sigma \rightarrow \mathbf{P}((e \rightarrow \iota \rightarrow t) \times \sigma)$.

$$(11) \quad \lambda s. \begin{cases} \langle \lambda x, i. \text{dryness}_{abs.}(w_i)(x) \geq \mathbf{d}_{dry\ abs.}(s), s \rangle & \tau_{dry\ abs.}(s) \\ \langle \lambda x, i. \text{dryness}_{rel.}(w_i)(x) \geq \mathbf{d}_{dry\ rel.}(c_i), s \rangle & \neg \tau_{dry\ abs.}(s) \end{cases}$$

Given some input state s , *dry* returns one of two possible senses—either the relative sense or the absolute sense—as according to a truth value $\tau_{dry\ abs.}(s)$ that depends on s . In case $\tau_{dry\ abs.}(s) = \mathbf{T}$, the adjective takes on the absolute sense and is true of an entity at an index i of the common ground just in case the entity’s dryness at i exceeds some globally determined threshold $\mathbf{d}_{dry\ abs.}(s)$ which depends on the state. If the adjective takes on the relative sense, it is true of an entity at i if the entity’s dryness at i exceeds a threshold which *also* depends on i . The idea is that there can be unresolved indeterminacy about the threshold of the adjective when this adjective is understood in its relative sense, and thus as vague; such uncertainty is usefully viewed as an aspect of the adjective’s core meaning, since it is associated with variability in how this meaning, when predicated of a particular entity, is true or false at different indices of the common ground. When the adjective is understood in its absolute sense, the indeterminacy about the threshold is instead *resolved* indeterminacy. Thus when its meaning is predicated of an entity, it may be true or false at an index of the common

ground depending *only* on the properties of this entity—*not* variation in the degree threshold.¹¹

To render such meanings in a somewhat more familiar format—i.e., one resembling a Bayesian network (Pearl 1985)—we introduce a *directed acyclic graph* (DAG) notation.¹² We can give an alternative representation of (11) as a DAG, as in (12).



The idea is that each node of the DAG in (12), except for the source node of the edge labeled s , represents a probability distribution—here, some degenerate distribution formed from a use of return. An edge going from a parent node to its child node represents how one probability distribution (the child) depends on the value sampled from another (the parent). An edge is labeled with the random variable whose value is sampled to determine the distribution represented by its child node. For example, the values of both m and n are sampled from degenerate distributions—ones which return the values specified in the boxes their edges lead out of. As a result, the values of m and n are *fixed*.

Importantly, there is an unlabeled root of the dag in (12), representing that the entire probability distribution depends on an input state s (which, in (11), is encoded as a λ -abstraction).

One useful aspect of this notation is its convenient representation of binding one probabilistic program to another. In general, given a DAG T_1 and a DAG

$$\begin{array}{c}
 \downarrow x \\
 T_2
 \end{array}$$

if T_1 represents a distribution over things of type α and the variable x is of type α , then we may form a third DAG:

$$\begin{array}{c}
 T_1 \\
 \downarrow x \\
 T_2
 \end{array}$$

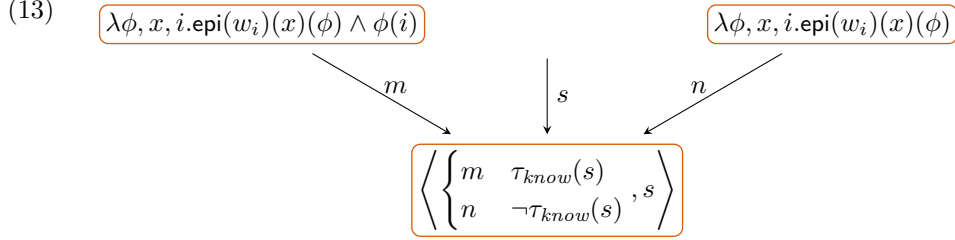
¹¹We consider the question of how to properly analyze the relative/absolute distinction to be open in principle; the foregoing discussion provides one particular encoding of it.

¹²The current notation may be understood as a Bayesian network which has each of its edges labeled by the variable sampled along that edge.

This DAG is understood to represent the result of concatenating the second DAG to the single terminal node of T_1 . The intended meaning is that x is sampled from T_1 before being used to compute the probability distribution T_2 , thus registering a new dependency.

3.7 The meaning of factivity

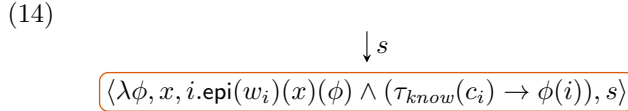
According to the Fundamental Discreteness Hypothesis, clause-embedding predicates entail the complement clauses they select with some probability in $[0, 1]$.¹³ For example, one can represent the meaning of *know* as in (13).



Here τ_{know} —analogous to $\tau_{dry\ abs.}$ above—determines whether to instantiate the meaning of *know* with a factive or a non-factive meaning. If $\tau_{know}(s)$ is **T**, *know* uses the interpretation m , which (a) ensures that the agent x is in the appropriate epistemic relation with the proposition ϕ denoted by the complement clause (whatever that may be), and (b) ensures that ϕ also holds at the index of evaluation. If $\tau_{know}(s)$ is **F**, the predicate uses the interpretation n , which imposes only the former requirement.

This scheme holds for all clause-embedding predicates. Further, variation among predicates is ultimately determined by the distribution over starting states for a given discourse. This prior distribution encodes one’s core lexical knowledge, as well as other kinds of knowledge, e.g., about what the starting common ground is likely to be.

Importantly, the lexical entry in (13) renders *know* semantically ambiguous. Ambiguity, on our account, is a potential source of resolved indeterminacy, since ambiguities are resolved on particular occasions of interpretation. The fact that an ambiguity is at stake is apparent in the shape of the DAG: two parent nodes of the terminal node represent, respectively, the two possible meaning for *know*. An alternative lexical entry rendering this indeterminacy as instead *unresolved* is given in (14).



According to (14), whether or not the complement clause of the predicate is entailed depends on the index of evaluation: τ_{know} —which now has a different type!—determines a truth value based on *this index*, and, if this truth value is **T**, the meaning of the complement is also required to be **T**.

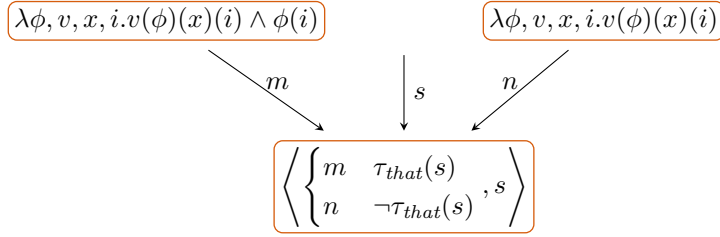
¹³We do not distinguish between factivity and veridicality for current purposes. This choice bears a resemblance to the general approach in Simons 2007.

It is important to point out that this characterization of factivity doesn't address the general factors distinguishing different predicates' interpretations. At the same time, we may say exactly how such factors ought to be encoded in our system: they pertain to the shape of the state s which is read in by either of the above possible meaning representations for *know*, and in particular (a) the value that s sets for $\tau_{\text{know}}(s)$ in (13), or alternatively (b) the value it sets for the common ground and the probability distribution over indices that this value encodes, which in turn determines the probability distribution over values for $\tau_{\text{know}}(c_i)$ in (14).

As a result, we can characterize the behavior of the predicates we investigate at a level of abstraction that remains uncommitted to any particular view about the factors controlling factivity, but which may be used to circumscribe more detailed accounts. We wish only to determine whether the gradience exhibited by clause-embedding predicates manifests unresolved indeterminacy (supporting the Fundamental Gradience Hypothesis) or resolved indeterminacy (supporting the Fundamental Discreteness Hypothesis). Thus we introduce only enough detail to distinguish these hypotheses without also determining what causal factors might lead to one or the other being true.

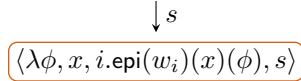
Note that alternatively, we could have described factivity as arising from a source external to the predicate; for example, by assuming that it is encoded by a complementizer (Kiparsky and Kiparsky 1970 *et seq.*).

$$(15) \quad \llbracket \textit{that} \rrbracket =$$



Given the alternative in (15), we need not assume that clause-embedding predicates, such as *know*, themselves give rise to factive interpretations.¹⁴

$$(16) \quad \llbracket \textit{know} \rrbracket =$$



This sort of approach in turn requires that contexts provide information about which predicate a given complementizer co-occurs with, since such information is needed in order for the probability of projection to be modulated by predicate type (see Gordon

¹⁴The implementation according to which the semantic value of the complementizer operates on the semantic value of the predicate is analogous to some neo-Davidsonian approaches to the semantics of propositional attitude verbs and complementizers (see Kratzer 2006; Moulton 2009; Bogal-Allbritten 2016; White and Rawlins 2018a) and might be adapted to approaches that regard the embedded clause as denoting an intersective modifier of eventualities (Elliott 2016; Elliott 2020).

and Chafetz 1990; Trueswell, Tanenhaus, and Kello 1993; MacDonald, Pearlmutter, and Seidenberg 1994; Garnsey et al. 1997; Altmann and Kamide 1999, i.a.). If the state makes this information available, then the implementation represented by (15) and (16) can yield the same range of statistical models as one according to which factivity is driven principally by the lexical entries for individual predicates.

4 Modeling

To investigate the analyses of factivity and world knowledge which are possible under the framework described in Section 3, we implement Bayesian models in the Stan programming language (Stan Development Team 2024) via the `CmdStanR` interface (Gabry and Češnovar 2023). We fit these models using Degen and Tonhauser’s (2021) experimental data and then compare them in terms of their expected log pointwise predictive densities (ELPDs) computed under the widely applicable information criterion (WAIC; Watanabe 2013; Gelman, Hwang, and Vehtari 2014), as implemented in R’s `loo` package (Vehtari et al. 2023). This measure quantifies how well each model fits the data, while also penalizing each model for how complex it is (i.e., the effective number of parameters it uses to fit the data).

In Section 4.1, we formalize our assumptions about the link between higher-order probabilistic programs of the kind described in Section 3 and participants’ response behavior. In Section 4.2, we describe and compare the two models which we fit to Degen and Tonhauser’s norming experiment data: a gradient model and a discrete model. We then describe each of the models of factivity possible under our framework in Section 4.3 before reporting our comparisons of these models, given Degen and Tonhauser’s projection experiment data, in Section 4.4.

4.1 Linking to response behavior

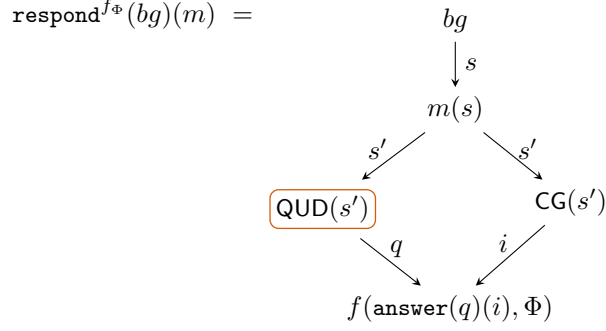
To connect the probabilistic programs associated with sentences to actual data, we need linking assumptions that relate these programs to the inference judgments that experimental participants report on a slider scale going between *no* and *yes* (cf. Jasbi, Waldon, and Degen 2019; Waldon and Degen 2020).

We assume that a slider scale can take on values in the closed unit interval $[0, 1]$. Specifying linking assumptions thus requires saying how probabilistic programs may be used to compute distributions on this interval.

4.1.1 Response models in the abstract

Here, we go into more detail about the response functions introduced in Section 3.5. Recall that response functions take two arguments: first, a probabilistic program bg of type $P\sigma$ representing a prior distribution over discourse states, and second, an ongoing discourse (m , of type $\sigma \rightarrow P(Qlr\sigma')$). Given these two arguments, a response function associates them with a response distribution determined by the current QUD:

$$\text{respond}^{f_\Phi: r \rightarrow P\rho} : P\sigma \rightarrow (\sigma \rightarrow P(Qlr\sigma')) \rightarrow P\rho$$

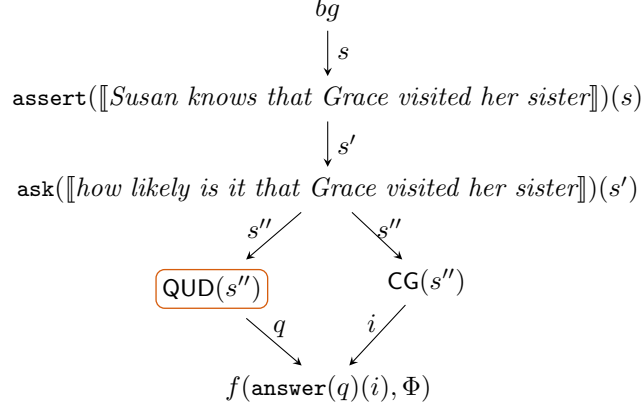


Going from top to bottom, a response is computed by: (a) sampling a state s from the prior distribution bg ; (b) applying the discourse m to s to obtain a distribution over outputs states; (c) sampling a new state s' from this distribution and grabbing the QUD from s' (along the left branch), as well as an index from the common ground of s' (along the right branch); and finally (d) computing an answer using the QUD and the index and feeding it to the response instrument. Note that the terminal node of this DAG has two parent nodes—given by the QUD (bound to q) and an index of the common ground (bound to i)—in order to encode the flow of information from both of these sources. Further, since we assume that questions denote sets of degree answers, **answer** is easily defined to take the maximum degree in a given set, at a given index:

$$\text{answer}(q)(i) = \max(\lambda d. q(d)(i))$$

This model of response behavior assumes that participants compute the probability of an answer to a question prompt by determining whether or not the answer is true at a random index of the common ground, weighted by how likely that index is, given the prior bg updated with the discourse m . They then attempt to respond with this answer, i.e., by mapping the chosen degree d onto a physical response of type ρ (whatever that may be). But due to factors independent of the process by which d is computed (e.g., inaccuracies in their ability to perfectly target the response corresponding to d on the response instrument), they produce an actual response according to the probability distribution $f(d, \Phi)$ for some fixed family of distributions $f : r \times p_1 \times \dots \times p_n \rightarrow Pr$ and vector $\Phi : p_1 \times \dots \times p_n$ of nuisance parameters.

For a schematic example, assume some prior bg of type $P\sigma$. Further, assume that the response instrument is a slider scale and can thus be modeled by the type r of real numbers, which we take to range in the interval $[0, 1]$. Then the following program of type Pr characterizes the distribution of slider responses in this interval. The idea here is that a response reflects a judgment about the likelihood that *Grace visited her sister* is true, given the information encoded by *Susan knows that Grace visited her sister*:



A suitable prior together with meanings for expressions like *likely*, along with bit of algebraic reasoning (see Grove and White 2025b), allows one to rewrite this program as the following one, assuming the *resolved* meaning for *know* of (13):

$$\begin{array}{c}
bg \\
\downarrow s \\
\left\{ \begin{array}{ll} f(1, \Phi) & \tau_{\text{know}}(s) \\ f(\text{likely}(\text{CG}(s))(\text{grace}), \Phi) & \neg \tau_{\text{know}}(s) \end{array} \right.
\end{array}$$

We employ a couple of abbreviations here to save space. $\text{grace} : \iota \rightarrow t$ denotes the proposition that Grace visited her sister. We also assume the existence of a function

$$\text{likely} : P\iota \rightarrow (\iota \rightarrow t) \rightarrow r$$

such that $\text{likely}(cg)(\phi)$ is the probability of the proposition ϕ , given the distribution over indices encoded by the common ground cg .

4.1.2 Examples of response models

This general set-up will remain invariant under the models considered in the rest of this section. What varies is the lexical entries we associate with clause-embedding predicates like *know*, along with the meanings of their complement clauses; crucially, whether they are assumed to represent factivity and world knowledge as resolved or unresolved sources of uncertainty.

The remaining challenge in implementing our proposals is to find a family of distributions over $[0, 1]$ —the set of values we use to encode bounded slider scales—that can flexibly capture a variety of possible response distributions. Finding such a family is challenging insofar as many of the families of bounded distributions commonly used in generalized linear mixed models, such as the beta or logit-normal distributions, only have support on the open interval $(0, 1)$; i.e., beta and logit-normal random

variables never take on 0 or 1 values. Because of this, they cannot describe the full distribution of slider responses, and in particular, the endpoints of the scale. This fact is particularly problematic in the current context, where endpoint responses are meaningful by hypothesis: consider the fact that according to the probability distribution defined above, the answer which is fed to the likelihood function f_Φ is 1 when *know* is understood as factive.¹⁵

We consider two approaches to this challenge. The first is *truncation*, whereby we model distributions on the closed unit interval by *truncating* a distribution on r . The second is *inflation*, whereby we model 0 and 1 responses as arising from a response process that mixes a distribution on $(0, 1)$ with a distribution on $\{0, 1\}$, thereby *inflating* the number of 0’s and 1’s relative to that expected under the distribution on $(0, 1)$. To implement the truncation approach, we assume f to be the family of truncated normal distributions $\mathcal{N}(\cdot) \mathbb{T}[0, 1]$, with nuisance parameter $\Phi \equiv \sigma$. To implement the inflation approach, we assume f to be the family of ordered beta distributions $\text{OrdBeta}(\cdot)$, with nuisance parameters $\Phi \equiv \langle \phi, \mathbf{c} \rangle$ (Kubinec 2023). Ultimately, we find that our main results are insensitive to which approach we take, so we focus on models implementing the truncation approach here (see Supplementary Materials 2 and 3.2 for discussions of the inflation models).

4.2 Models of the norming data

To construct our models, we use a *pipelined* approach:¹⁶ we first fit models of Degen and Tonhauser’s norming data (experiment 2a) and obtain posterior distributions for parameters associated with the forty pairs of complement clauses and facts (what we refer to as *contexts*). We then used these posterior distributions as prior distributions for the corresponding parameters in the different models of Degen and Tonhauser’s projection experiment data (experiment 2b) implied by our framework. We present these models in Section 4.3.

Recall that the prompts that participants were presented in the norming task asked about the likelihood of a fact related to world knowledge, given some other background fact. To model the resulting inference judgments, we fit two kinds of models: (i) a *gradient* model, according to which inferences conditioned by world knowledge are driven by unresolved indeterminacy; and (ii) a *discrete* model, according to which such inferences are driven by resolved indeterminacy. As we’ll see, the gradient model of the norming data provides the better fit (see Section 4.2.3). We therefore use this model to extract prior distributions for the parameters of the models of factivity.

4.2.1 World knowledge as a gradient phenomenon

The gradient model of the norming data encodes uncertainty about world knowledge in terms of unresolved indeterminacy. We give it schematically as follows, again calling the prior distribution ‘*bg*’, and assuming—following the question prompt used in Degen

¹⁵A common approach used in regression analyses is to sidestep this issue by “nudging” 0 and 1 responses toward 0.5 by a small δ , thereby putting them into $(0, 1)$. This approach is itself problematic because it introduces a researcher degree-of-freedom; moreover, it is especially problematic in the current context for reasons which we discuss in Supplementary Materials 1.

¹⁶For further details, see Supplementary Materials 3.

and Tonhauser’s task—that the response is a function of an answer to the question *how likely is it that S* :

$$\begin{array}{lcl}
 \text{norming-gradient} & = & bg \\
 & & \downarrow s \\
 & & \llbracket S \rrbracket(s) \\
 & & \downarrow \langle \phi, s' \rangle \\
 & & \boxed{\text{likely}(\text{CG}(s'))(\phi)} \\
 & & \downarrow d \\
 & & f(d, \Phi)
 \end{array}$$

4.2.2 World knowledge as a discrete phenomenon

The discrete model of the norming data encodes uncertainty about world knowledge in terms of resolved indeterminacy. We give it as follows:

$$\begin{array}{lcl}
 \text{norming-discrete} & = & bg \\
 & & \downarrow s \\
 & & \llbracket S \rrbracket(s) \\
 & & \downarrow \langle \phi, s' \rangle \\
 & & \boxed{\text{likely}(\text{CG}(s'))(\phi)} \\
 & & \downarrow d \\
 & & \text{Bernoulli}(d) \\
 & & \downarrow b \\
 & & \begin{cases} f(1, \Phi) & b \\ f(0, \Phi) & \neg b \end{cases}
 \end{array}$$

The difference between this model and the gradient model is the continuation of the likelihood value with a Bernoulli program, which returns **T** with probability corresponding to that likelihood and **F** with a probability of one minus that likelihood. That is, the answer to the *how likely* question has been discretized, so that it is now 1 with probability determined by the original answer (and 0 otherwise). This model might be understood as assuming that world knowledge is tied directly to the *discourse state*, rather than the common ground: instead of computing the truth of S relative to the common ground of the state s' in order to determine its likelihood, it extracts an index of evaluation from s' immediately ($i(s')$). It then computes a likelihood of either 1 or 0, depending on whether S is true or false at that index:

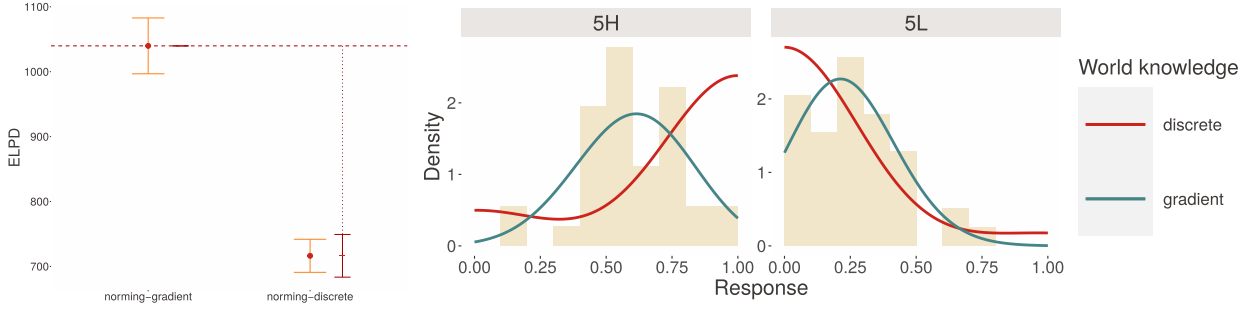


Fig. 5 Left: ELPDs for the two models of the norming data. Dotted line indicates estimated difference between the norming-discrete model and the norming-gradient model. Error bars indicate standard errors. Right: posterior predictive distributions for both models for the item *Sophia got a tattoo*, given either the fact *Sophia is a hipster* (5H) or the fact *Sophia is a fashion model* (5L).

$$\begin{array}{c}
bg \\
\downarrow s \\
\llbracket S \rrbracket(s) \\
\downarrow \langle \phi, s' \rangle \\
\boxed{\text{likely}(\text{CG}(s'))(\lambda i. \phi(i(s')))} \\
\downarrow d \\
f(d, \Phi)
\end{array}$$

4.2.3 Comparison

The ELPDs for the two models of the norming data are shown in Figure 5 (left plot). It can be seen that the gradient model substantially outperforms the discrete model, indicating that responders represent their uncertainty about world knowledge gradiently.

To give a sense of why the gradient model performs so much better, Figure 5 also shows the posterior predictive distributions for the two models for two items (right plot; see Figure 6 of Supplementary Materials 4.1 for all items). These plots provide a visual indication of how well each model fits the distribution of responses to the norming items: the closer a particular curve is to the shape of the histogram, the better the corresponding model fits the data. Indeed, the gradient model provides a superior fit due its ability to capture the fact that responses often cluster away from the endpoints of the scale. Meanwhile, the discrete model only allows the theoretical response (i.e., before any error is added) to occur at a scale endpoint. This limitation often causes the posterior predictive of the discrete model to have a bimodal distribution, as the plot for 5H illustrates.

4.3 Models of factivity

We now provide our models of factivity and prior world knowledge, which we fit to Degen and Tonhauser’s projection experiment data.¹⁷ Each of these models assumes that factivity provides either a discrete or a gradient contribution to inference. The models we present here make these assumptions wholesale, either regarding every predicate as making a discrete contribution or regarding every predicate as making a gradient contribution. Our current purpose is to compare these strong versions of the two hypotheses of Section 2, but we investigate the interesting possibility that predicates vary in their discreteness in ongoing work.

4.3.1 Factivity as a fundamentally discrete phenomenon

The first theory we consider regards sentences containing clause-embedding predicates as either triggering or not triggering factive inferences on particular occasions of use, depending on either (i) the interpretation of (a) the clause-embedding predicate, or (b) the surrounding structure; or (ii) some discourse structure, such as the common ground or question under discussion, whose properties are conditioned on pragmatic knowledge about the sentence. The uncertainty about whether or not a given predicate’s complement clause projects is thus regarded as arising from resolved indeterminacy.

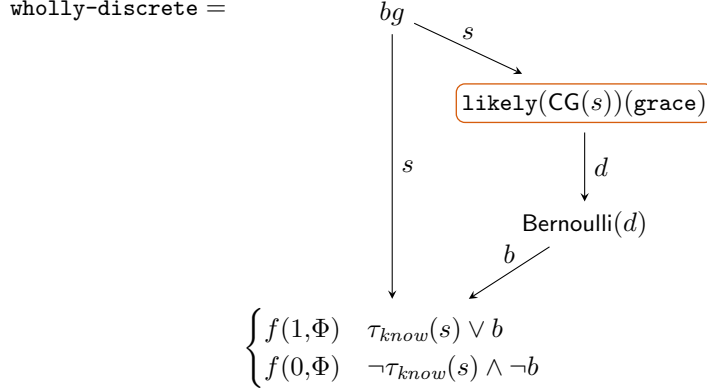
Assuming that factivity is a discrete phenomenon, the question remains how world knowledge affects inferences about the truth of an embedded clause. Here two possibilities arise. The first allows for uncertainty related to world knowledge to manifest as unresolved indeterminacy, which may, in turn, make individual judgments of truth uncertain. We refer to this theory as the *discrete-factivity* theory, emphasizing that it regards the contribution factivity makes as fundamentally discrete in nature. It gives rise to the model illustrated in Section 4.1.1, reproduced here again for the predicate *know* and the complement clause *Grace visited her sister*.¹⁸

$$\text{discrete-factivity} = \begin{array}{c} bg \\ \downarrow s \\ \left\{ \begin{array}{ll} f(1, \Phi) & \tau_{know}(s) \\ f(\text{likely}(\text{CG}(s))(\text{grace}), \Phi) & \neg \tau_{know}(s) \end{array} \right. \end{array}$$

¹⁷See Supplementary Materials 3 for the full model specifications.

¹⁸Each of our models of factivity encodes the QUD as a *how likely* question (e.g., *how likely is it that Grace visited her sister*). This choice of QUD—which we take to approximate the actual polar questions Degen and Tonhauser use in their experiments—yields the required sets of degrees. See Grove and White 2025a, which shows that collecting data under a task featuring a degree question tends to improve overall fit for models that appropriately account for the semantics of the question prompt.

The second possibility is that both factivity *and* world knowledge are discrete in nature.



This *wholly-discrete* model effectively hypothesizes that no inferences display uncertainty on particular occasions of use.¹⁹ Any gradience displayed in people’s measured inferences must therefore be due to task effects, as modeled by f_Φ .

Note that the discrete-factivity model effectively provides a more complete version of the norming-gradient model described in Section 4.2, whereas the wholly-discrete model effectively completes the **norming-discrete** model. Hence, we might have an *a priori* expectation that between the two, the discrete-factivity model should provide a better fit to the data, mirroring the superiority of the norming-gradient model to the norming-discrete model.

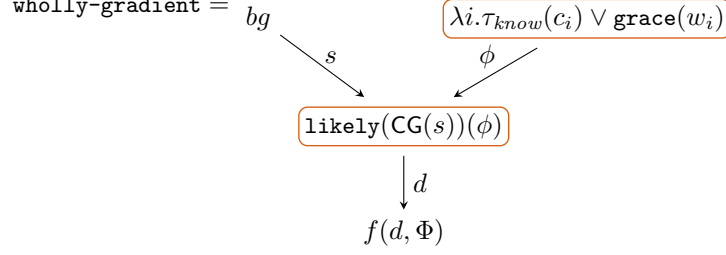
4.3.2 Factivity as a fundamentally gradient phenomenon

The theory that regards factivity as fundamentally gradient in nature does so by pushing what would otherwise be resolved indeterminacy about factivity onto the unresolved indeterminacy level. This theory also has two possible implementations that vary with respect to whether world knowledge is modeled as token-level indeterminate or type-level indeterminate. We refer to the version of the theory that regards both factivity and world knowledge as token-level indeterminate as the *wholly-gradient* version.

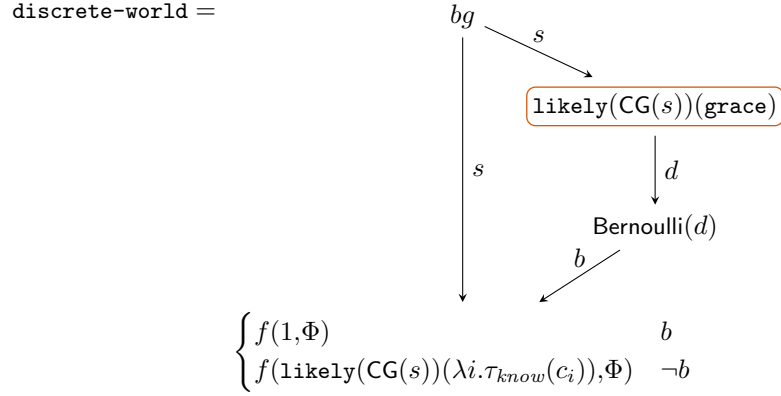
We obtain the corresponding model by modifying the discrete-factivity model so that it is effectively constructed from the lexical entry for *know* in (14), rather than

¹⁹We don’t take anyone to endorse such a model, and the results from our fits to the norming data speak against it.

the one in (13):



We refer to the alternative version of the model that regards world knowledge as discrete and factivity as gradient as the *discrete-world* version. As with the wholly-discrete model, we do not take anyone’s theory of factivity to be consistent with the discrete-world model, in particular, but we include it for completeness. This model can be seen as the mirror image of the discrete-factivity model:



Among the four models we consider, we take the wholly-gradient model to come the closest to implementing the Gradient Projection Principle (Tonhauser, Beaver, and Degen 2018). This model effectively provides an alternative completion of the norming-gradient model described in Section 4.2; and the discrete-world model provides an alternative completion of the norming-discrete model.

4.4 Comparisons

Figure 6 (left plot) provides ELPDs estimated for the four models, based on log-likelihoods computed from Degen and Tonhauser’s experimental data. We observe that the discrete-factivity model captures the data the best, while the wholly-discrete model trails behind it. The wholly-gradient and discrete-world models—the two that assume factivity to be gradient—perform the worst.

We therefore have preliminary evidence that the best model of data collected using Degen and Tonhauser’s task treats factive presupposition projection as a discrete

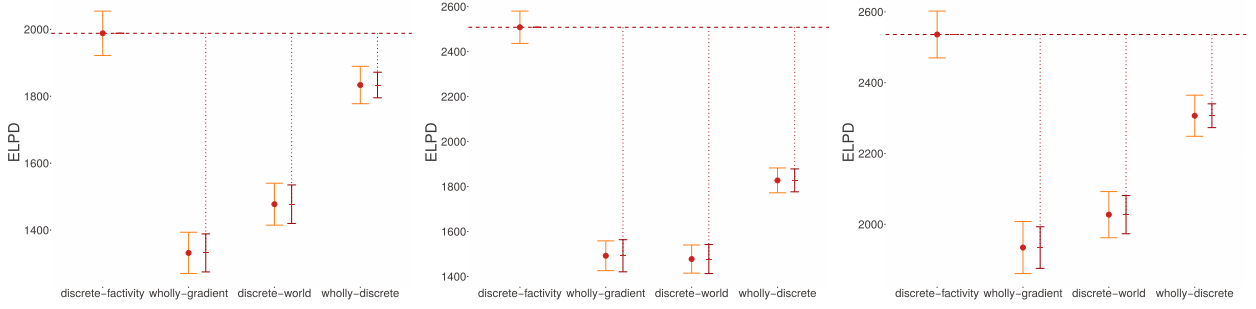


Fig. 6 Left: ELPDs for the four models. Center: ELPDs for the four models with an anti-veridicality component added for each predicate. Right: ELPDs for the four model evaluations on our replication experiment data. Dotted lines indicate estimated differences between each model and the discrete-factivity model. Error bars indicate standard errors.

phenomenon and the inferences contributed by world knowledge as gradient. Further, note that by simply modifying the discrete-factivity model so that it treats factivity as gradient, one goes from the *best*-performing model to the *worst*-performing one.

To give a sense of the performance of the models as assessed against the inference judgment data, the posterior predictive distributions for each model are plotted for six predicates, for all contexts combined, in Figure 7 (see Figure 7 of Supplementary Materials 4.1 for all predicates). We observe here that the models that assume that either factivity or world knowledge is discrete are better able to capture dips in the frequency of responses in the middle of the scale than the wholly-gradient model. Indeed, the wholly-gradient model predicts distributions that are much smoother than do the models that assume some form of discreteness. This behavior is the main reason the wholly-gradient model fits the inference judgment data the worst of any of the models.

Models that assume some amount of discreteness capture the dips in frequency toward the middle of the scale, in part, because they effectively regard the distribution over inference judgments as a mixture of distributions: at least one with a mode at 1 and another with a mode determined by the structure of the particular model. In contrast, the wholly-gradient model only assumes a single distribution.²⁰ This property of the models that assume some amount of discreteness makes them more complex than the wholly-gradient model—in the sense that they have more effective parameters—but as the pattern of ELPDs in Figure 6 shows, this additional complexity is offset by how much better they fit the data. (Recall that ELPD explicitly quantifies how well a model fits the data while penalizing model complexity, as measured by the effective number of parameters.)

Notably, none of the four models fits the data perfectly. For instance, the canonically non-projective predicates *think* and *pretend* have distributions which all four of our models appear to have difficulty capturing, at least by visual inspection (see

²⁰Multi-modality may still arise in the posterior predictive distributions for the wholly-gradient model, but it can only arise due to participant random effects (see Supplementary Materials 3 for formal details) or differences between items for a given predicate (see, e.g., *confirm* in Figure 7).

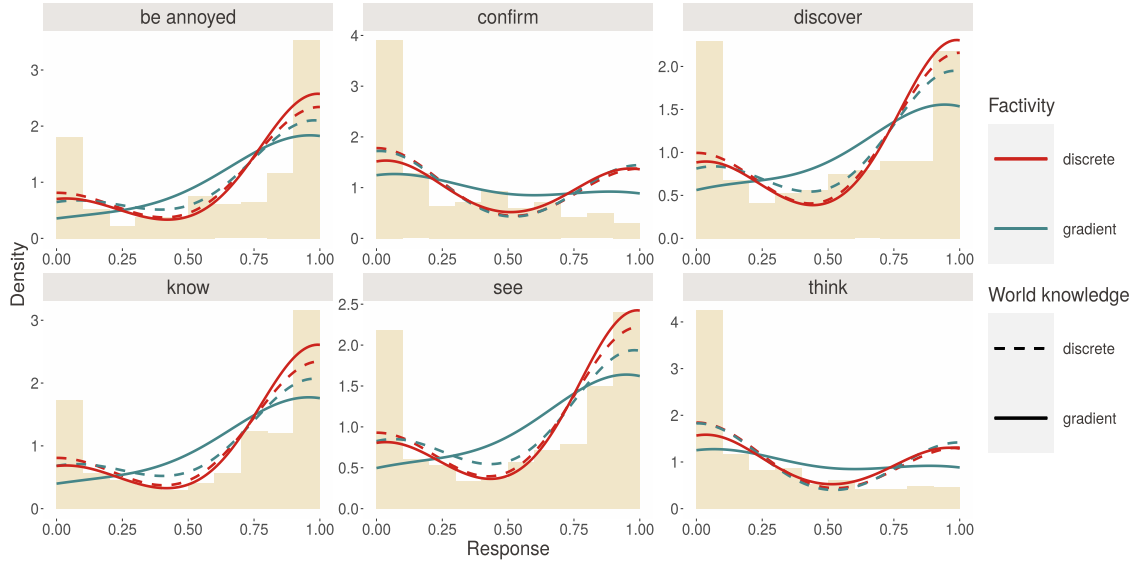


Fig. 7 Posterior predictive distributions (with simulated participant intercepts) of all four models for six predicates from Degen and Tonhauser’s (2021) projection experiment, for all contexts combined. Empirical distributions are represented by density histograms of data from Degen and Tonhauser 2021.

Figure 7 of Supplementary Materials 4.1). This difficulty appears to be due to an anti-veridicality inference associated with these predicates—an inference that the content of the complement clause is *not* true.²¹

Since we consider denotations for predicates that vary only with respect to veridicality and non-veridicality—not anti-veridicality—no model captures the latter. To assume a three-way distinction among veridicality, non-veridicality, and anti-veridicality would allow the discrete-factivity and wholly-discrete models to mix in another distribution with a mode at 0, while also allowing the wholly-gradient model an extra parameter to modulate the gradient associated with individual predicates.

To investigate the possibility that such a modification affects the model ranking, we also fit models which add an anti-veridicality component for each predicate (see Supplementary Materials 3.1.2). We find that, while the anti-veridicality component substantially improves the performance of the discrete-factivity model (as illustrated by the plots in Figure 8 of Supplementary Materials 4.1) the only change in relative model performance is that the wholly-gradient model now slightly outperforms the discrete-world model, as shown in Figure 6 (center plot).

²¹These inferences likely arise from different sources; e.g., the lexical semantics of *pretend*, but a conversational implicature in the case of *think*. More generally, inferences of this kind may in principle arise for *any* predicate on its non-factive interpretation, if factivity is discrete, which may shed additional light on the mass of responses at or close to 0 in the empirical data across predicates.

5 Evaluations

Strictly speaking, the comparisons we report in [Section 4](#) are *post hoc*; and while the results are suggestive, we cannot draw firm conclusions without a replication of Degen and Tonhauser’s experiment. We report such a replication in [Section 5.1](#) and find the same pattern of model comparison results: the models that assume that factivity is discrete reliably outperform those that assume it is gradient. To ensure that these results are not somehow driven by the particular discourse contexts used in Degen and Tonhauser’s experiments, we collect two additional datasets that use the same paradigm, but that mask the contents of the embedded clause in two different ways. In these experiments, which we report in [Section 5.2](#), we again find that the discrete-factivity models outperform all of the other models.

5.1 Experiment 1: held-out projection experiment data

5.1.1 Materials

Our materials and methods were identical to those of Degen and Tonhauser ([2021](#)).

5.1.2 Participants

We collected data from 300 participants using Amazon Mechanical Turk, paying each participant one dollar. Each participant was required to pass the qualification test described in White, Hacquard, and Lidz [2018](#), in order to ensure that they were a native speaker of American English. We removed data from two participants who claimed to have technical difficulties completing the experiment, and from ten more whose performance was more than two standard deviations below the mean on the six control items, leaving us with data from a total of 288 participants.

5.1.3 Results

[Figure 8](#) shows the item means from our replication experiment plotted against Degen and Tonhauser’s original experiment, along with the means obtained by collapsing across contexts. We observe that there is strong agreement (Spearman’s $r = 0.98, p \leq 0.001$) between these means and those obtained from Degen and Tonhauser’s data.

5.1.4 Model fitting

To evaluate the four models using this data, we obtain, from each model, means μ_ν and standard deviations σ_ν of the marginal posterior distribution of the log-odds of projection for each predicate, as well as means μ_ω and standard deviations σ_ω of the marginal posterior distribution of the log-odds certainty for each context. We then use normal distributions with these means and standard deviations as prior distributions for the corresponding parameters in the models constructed for the evaluations.²²

²²See Supplementary Materials 3.1.3 for further details concerning these models.

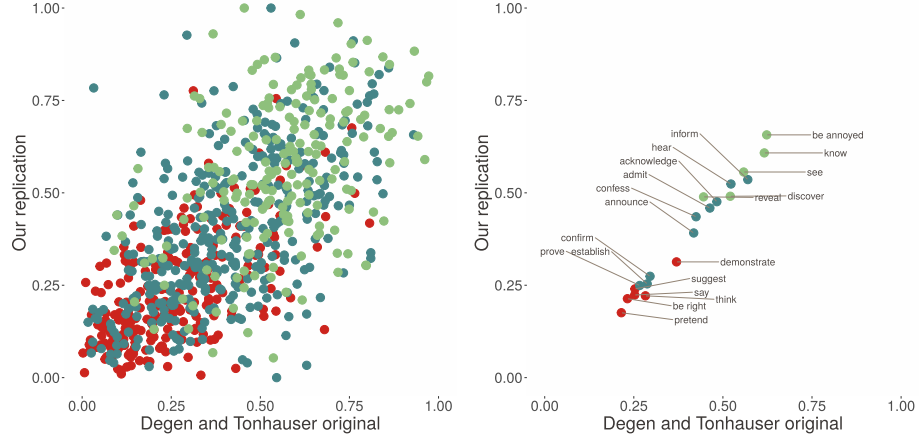


Fig. 8 Degen and Tonhauser’s (2021) projection data versus our replication. Left: item means (Spearman’s $r = 0.68, p \leq 0.001$). Right: verb means (Spearman’s $r = 0.98, p \leq 0.001$). For both, “non-factive” verbs are in red, “optionally factive” verbs are in teal, and “canonically factive” verbs are in green.

5.1.5 Model comparison

Figure 6 (right plot) provides ELPDs estimated for the four model evaluations, based on log-likelihoods computed from the data obtained in our replication experiment. The pattern of goodness-of-fit observed here is extremely close to that exhibited by the original model comparison on the left in Figure 6: the discrete-factivity model fares the best, followed by the wholly-discrete model. The wholly-gradient and discrete-world models fare the worst.

We therefore have evidence that the differences in model performance reported in the previous section are quite robust, at least when assessed using data from Degen and Tonhauser’s experimental task. The next two experiments provide a test of the models in a somewhat different setting—one in which the uncertainty contributed by the linguistic contexts of the predicates of interest is sent to an extreme.

5.2 Experiments 2 and 3: non-contentful contexts

We now test the inferred posterior distributions over probabilities of projection on held-out data from two experiments in which the context of each predicate is stripped of the rich lexical content that partially governs the inferences produced in the original experiment. We obtain contexts for this evaluation in two ways. In our Experiment 2, we *bleach* each predicate’s complement clause so that it is just *a particular thing happened*. In Experiment 3, we use a *templatic* complement clause of the form *X happened*.

These manipulations serve two purposes. First, they allow us to assess the performance of the four models when the source of variance among inference judgments contributed by prior world knowledge is removed. Second, they put the predicates in environments in which knowledge about the context is minimal; as a result, they may produce greater uncertainty in people’s inferences. This additional uncertainty

could confer an *a priori* advantage to the wholly-gradient model, which considers all inferences, even those triggered by projective predicates, to be beset with uncertainty. We therefore consider these evaluations to be a stronger test of the discrete-factivity model’s edge over the wholly-gradient model than the original experiments were.

5.2.1 Experiment 2: bleached items

To construct the bleached items, we placed each of the twenty predicates investigated in Degen and Tonhauser’s experiment in a context in which its subject was one of the proper names from the original experiment, and in which its complement clause was just *a particular thing happened*. On each trial, we provided participants with a background context that was intended to make the prompt as natural as possible. The only thing we varied in this background context from one trial to the next was the name of the individual who makes the relevant utterance. Finally (taking that individual’s name to be *P*), we prompted participants to answer the question *Is P certain that that thing happened?* on a sliding scale with *no* on the left and *yes* on the right. The following experimental trial is the one for the predicate *pretend*:

You are at a party. You walk into the kitchen and overhear Linda ask somebody else a question. Linda doesn’t know you and wants to be secretive, so speaks in somewhat coded language.

Linda asks: “Did Tim pretend that a particular thing happened?”

Is Linda certain that that thing happened?

no yes

Next

In addition to the twenty bleached items, participants saw six control items which were constructed in order to somehow incorporate a bleached subordinate clause; for example, *Did Madison have a baby, despite the fact that a particular thing happened?*. All six control items had an intended response of 1.

5.2.2 Experiment 3: templatic items

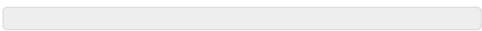
To construct the templatic items, we placed each of the same twenty predicates in a context in which its subject was, again, a proper name from the original Degen and Tonhauser experiment, and in which its complement clause was *X happened*. We provided a background context on each trial, so that the prompt was natural. Background contexts, again, only differed from one another in the name of the individual who makes the relevant utterance. Given a trial on which the individual *P* was the

speaker, we prompted participants with the question *Is P certain that X happened?*, which they answered on a sliding scale with *no* on the left and *yes* on the right. The following example trial is the one for the predicate *pretend*:

You are at a party. You walk into the kitchen and overhear William ask somebody else a question. The party is very noisy, and you only hear part of what is said. The part you don't hear is represented by the 'X'.

William asks: "*Did Ray pretend that X happened?*"

Is William certain that X happened?

no  yes

Next

Participants again saw six control items which were constructed in order to incorporate a templatic subordinate clause; for example, *Did Madison have a baby, despite the fact that X happened?*.

5.2.3 Participants

For each experiment, we collected data from 50 new participants using Amazon Mechanical Turk, paying each participant one dollar. Each participant was again required to pass the qualification test described in White, Hacquard, and Lidz 2018, and any participant whose average score on the control items did not fall within two standard deviations below the mean of all participant's responses was excluded from the analysis. Using this criterion, three participants' data was excluded from Experiment 2, leaving 47 participants for analysis; and one participant was excluded from Experiment 3, leaving 49 participants for analysis.

5.2.4 Results

We observe in Figure 9 that the responses elicited by both the bleached (Spearman's $r = 0.97, p \leq 0.001$) and templatic (Spearman's $r = 0.87, p \leq 0.001$) items track the gradient knowledge about factivity that people deploy in the typical contentful setting extremely well. Not only is the same type of gradience observed among predicates when they are placed in bleached or templatic contexts, but the rankings among predicates are maintained almost entirely.²³ This finding furthermore suggests that it

²³Notably, the range of average ratings for predicates in Experiment 3 is not as wide as exhibited in the previous experiments, with most falling between 0.5 and 0.75, suggesting that there may have been a great deal of uncertainty governing the inferences produced from this task.

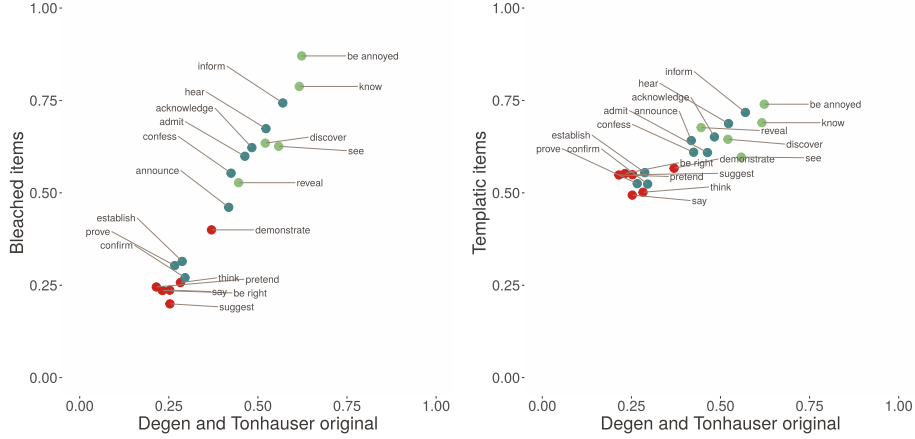


Fig. 9 Degen and Tonhauser’s (2021) projection data versus data from contexts with minimal lexical content. Left: bleached data (Spearman’s $r = 0.97, p \leq 0.001$). Right: templatic data (Spearman’s $r = 0.87, p \leq 0.001$). For both, “non-factive” verbs are in red, “optionally factive” verbs are in teal, and “canonically factive” verbs are in green.

is safe to compare modeling results obtained from only bleached or templatic items to those obtained from contentful items—e.g., the clustering results of Kane, Gantt, and White 2022 discussed in Section 2—at least if those results pertain to the aggregate responses for individual predicates.

5.2.5 Model fitting

To evaluate the four models using both the bleached and templatic data, we use the same means μ_ν and standard deviations σ_ν of the marginal posterior distributions of the log-odds of projection that we used for the evaluations on the replication experiment data. As before, we use normal distributions with these means and standard deviations as prior distributions for the corresponding parameters in the models constructed for the evaluations. Then, in each evaluation, we infer a distribution over the parameters σ_ω and ω that regulate the certainty associated with either the bleached or the templatic context.²⁴

5.2.6 Model comparison

Figure 10 provides ELPDs for all four models evaluated on Experiments 2 and 3. We see here that the discrete-factivity model fares the best in both experiments, while the other three models fare comparably with each other. It is somewhat remarkable that the more fine-grained differences among models observable from both the original fits and the evaluation on the replication data do not appear to hold up under the current evaluation. For example, the wholly-discrete model is no longer easily distinguished from the wholly-gradient and discrete-world models by its better performance. Rather, the discrete-factivity model seems to have a unique advantage.

²⁴See Supplementary Materials 3.1.4 for further details concerning these models.

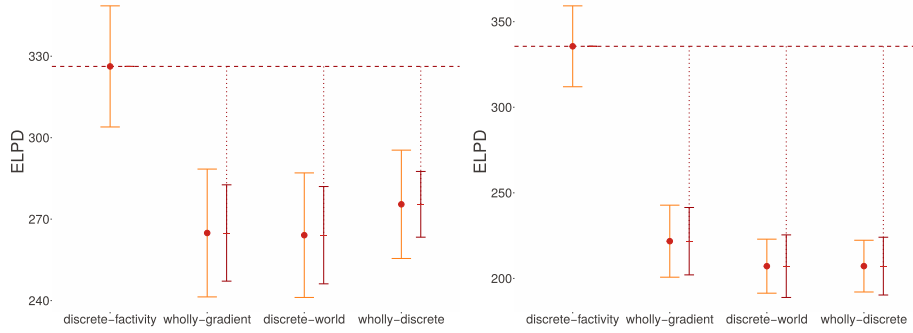


Fig. 10 ELPDs for the four model evaluations on the bleached data (left) and the templatic data (right). Dotted lines indicate estimated differences between each model and the discrete-factivity model. Error bars indicate standard errors.

This difference is especially notable for Experiment 3, given the somewhat squashed average responses seen across verbs in Figure 9. Despite the high amount of uncertainty which this task may have produced, such uncertainty seems to have been filtered through the discrete behavior of factive inferences. The uncertainty about such inferences appears to relate to the interpretation of a given predicate, rather than a gradient contribution which that predicate makes to an inference.

To give a sense of the differences among the model evaluations on the bleached and templatic data, Figure 9 and Figure 10 of Supplementary Materials 4.1 show the posterior predictive distributions of these evaluations for all predicates.

6 Consequences for theories of factivity

We have compared four models of the task reported in Degen and Tonhauser 2021, each fit to their experimental data. These models differ from one another along two axes. First, whether they consider the contribution of prior world knowledge to inferences about the truth of the clause embedded by a predicate to involve resolved or unresolved indeterminacy; and second, whether they consider the relevant predicate’s contribution to these inferences to involve resolved or unresolved indeterminacy.

Our initial comparison of the four models found that a model that associates world knowledge with unresolved indeterminacy and predicate knowledge with resolved indeterminacy best accounts for the distributions of judgments in Degen and Tonhauser’s experimental data. Follow-up evaluations of the four models confirmed this initial comparison: the same model best accounts for held-out data from a replication of Degen and Tonhauser’s experiment, for which distributions over the parameters of interest are extracted from the posteriors of the initial models (see Figure 6, right plot). The discrete-factivity model also best accounts for data from two tasks in which predicates are placed in contexts with minimal lexical content (see Figure 10).

Taken together, these results provide strong evidence that the gradience observed by White and Rawlins (2018b) and studied by Degen and Tonhauser is driven by *resolved* indeterminacy—it is associated with variation across experimental trials in

the *type* of an expression—and thus, whether or not it gives rise to a particular inference. This finding in turn suggests that clause-embedding predicates differ in the frequencies with which they trigger projective inferences, but the contribution a predicate makes on particular occasions of use (or interpretation) is *discrete*: either it produces the relevant inference, or it does not produce it at all. These results provide strong support for the Fundamental Discreteness Hypothesis, discussed in [Section 2](#). Here we say a bit more about what we think they say for theories of factivity—and presupposition projection more generally.

In [Section 2](#), we considered three ways in which the Fundamental Discreteness Hypothesis might be cashed out: (i) clause-embedding predicates may have multiple senses—at least one that is implicated in triggering projection and at least one that is not; (ii) clause-embedding predicates may occur in multiple structures—at least one that is implicated in triggering projection and at least one that is not; or (iii) clause-embedding predicates may be usable in contexts where the common ground—or some other construct, such as the question under discussion—entails the content of its embedded clause, as well as contexts where it does not (see Simons, Beaver, et al. [2017](#); Roberts and Simons [2024](#)). Under the first two explanations, which we may describe as *conventionalist*, the resolved indeterminacy we observe is ultimately related to lexical or structural ambiguity; under the third explanation, which we may describe as *conversationalist*, this indeterminacy pertains to what kind of discourse contexts an expression tends to occur in. Importantly, modern accounts of presupposition projection tend to *blend* conventionalist and conversationalist strategies in explaining particular phenomena, with most accounts leaning more heavily in one direction or the other (see White [2019](#) for an overview). We illustrate this fact in [Section 6.1](#) for one broadly conventionalist account of the projective inferences of emotive predicates (Djäv, Zehr, and Schwarz [2018](#)). In [Section 6.2](#), we briefly mention the conversationalist accounts of projection found in Simons, Beaver, et al. [2017](#) and Roberts and Simons [2024](#) which, taken together with our results, would require any indeterminacy about discourse constructs such as the QUD to be resolved in nature.

6.1 Conventionalist accounts

Consider an emotive predicate such as *be annoyed*. In addition to giving rise to projective inferences, such a predicate tends to produce inferences about the beliefs of the entity it is predicated of. For example, [\(17\)](#) tends to give rise to both the inferences in [\(18a\)](#) and in [\(18b\)](#).

(17) Was Jo annoyed that Mo left?

- (18) a. Jo believed that Mo left.
b. Mo left.

One species of conventionalist account attributes [\(18a\)](#) and [\(18b\)](#) to independent semantic properties of [\(17\)](#) (see Villalta [2008](#); Romero [2015](#); Uegaki and Sudo [2019](#), i.a.).

Another attributes both inferences to the same property, i.e., deriving one from the other via additional pragmatic assumptions (Djäv, Zehr, and Schwarz 2018). Adapting ideas put forth by Heim (1992, p. 212), for instance, (18b) might be derived from (18a) by invoking a defeasible assumption that Jo is credible. On this sort of account, (18a) and (18b) need not be triggered independently.

This second species of account can be extended to emotive factives as a class (Djäv, Zehr, and Schwarz 2018, p. 382), given that all emotive predicates tend to give rise to such belief inferences (see Kane, Gantt, and White 2022). Gradiance might then be a product of variability either in whether or not a belief inference is drawn or in whether or not an assumption of credibility is made.

6.2 Conversationalist accounts

Conversationalist approaches propose that knowledge of predicates like *be annoyed* relates those predicates to distributions over discourse-theoretic constructs, such as the common ground or the question under discussion. Such distributions are likely constrained by general pragmatic principles, which would, in turn, interact with conceptual knowledge associated with a given predicate to generate particular inferences (see Simons, Beaver, et al. 2017; Roberts and Simons 2024). Crucially, lexical semantic presuppositions are not the main driver of projective inferences under this sort of account—if they are implicated at all.

For instance, Simons, Beaver, et al. put forth a conversationalist account that relies on the notion of a QUD. On their account, whether or not the complement of a clause-embedding predicate projects varies according to prosodic and contextual factors conditioned by the QUD.²⁵ Specifically, Simons, Beaver, et al. argue that the projective inferences licensed at any point in a discourse are simply those which are backgrounded by the QUD, while the non-projective inferences are those which are made at-issue (see also Simons 2001; Simons 2007, i.a.). Simons, Beaver, et al. provide the example in (19) (their ex. 10) to illustrate.

- (19) A: James just found out that Harry’s having a graduation party, and I can’t understand why he’s so upset about it.
 B: He didn’t find out that HARRY’s having a graduation party...
 ... he found out that HARRIET is having a graduation party, and HARRIET is his best friend.

Crucially, while the content of the complement of *find out* projects in other contexts, it does not appear to project in (19) due to the presence of focus on the embedded subject *Harry*. On the assumption that *find out* does not lexically trigger a projective inference, it ought not to trigger one in the context of (19) either, according to Simons,

²⁵Importantly, lexical knowledge also conditions QUD choice under standard accounts in the Karttunen-Heim tradition; it just does so indirectly. If, following Heim (1983), a sentence denotes a context-change potential, and its presuppositions are encoded as definedness conditions, then it will only be defined on context sets that satisfy its presuppositions. If (following, e.g., Roberts (2012)) a QUD is just a partition of the context set, the update denoted by a sentence is defined just in case that sentence’s presuppositions are true in every cell of the partition—i.e., if the QUD entails the sentence’s presuppositions. As a result, variability in a sentence’s definedness conditions may be seen to covary with QUDs insofar as the latter background the inferences triggered by those definedness conditions.

Beaver, et al., under the assumption that the QUD consists of alternatives according to which James found out that someone or other is having a graduation party.

Roberts and Simons (2024) extend this idea by arguing that “projectivity... is predictable from the detailed lexical semantics of the triggers, in combination with broad pragmatic principles” (p. 2), but crucially also that “projective readings of [change of state] predicates, of factives, and of predicates that trigger selectional restrictions, are not conventionally derived” (p. 29). Rather, they hold that these projective inferences are driven by “the nuanced and probabilistic knowledge that language users have about the functions, effects and contexts of use of linguistic forms” (p. 27), and they work through various specific examples.

7 Conclusion

The results reported here can be understood as consistent with a view of factivity as a discrete property of an expression on a particular occasion of use. Sometimes, a predicate (depending on the predicate) gives rise to a projective inference, and sometimes it does not (again, depending on the predicate). It never partially projects. This result is in line with the view originally advocated by Kiparsky and Kiparsky (1970), Karttunen (1971), i.e., that certain predicates can be understood to discretely trigger a presupposition that the clause they select is true.

But while we address the nature of gradience at the token level, our results do not speak to the importantly distinct question of whether there exist classes of factive predicates at the type level. Alone, our results are consistent with a view whereon there are no semantic classes of factive predicates. But in conjunction with the results of Kane, Gantt, and White (2022) (see Section 2), who do find the existence of factive classes of predicates (pace Degen and Tonhauser 2022), a clear picture emerges: factivity displays discreteness at both the type and token levels, in line with the traditional view. A remaining question is how this discreteness is derived: by conventional means, by conversational means, or (most likely) by some mixture of the two (Roberts and Simons 2024).

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